IOP Publishing | London Mathematical Society Nonlinearity

Nonlinearity **38** (2025) 035001 (44pp) https://doi.org/10.1088/1361-6544/adac9c

Optimal linear response for expanding circle maps

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Received 27 January 2024; revised 27 December 2024
Accepted for publication 21 January 2025
Published 3 February 2025

Recommended by Dr Mark Pollicott

Abstract

We consider the problem of optimal linear response for deterministic expanding maps of the circle. To each infinitesimal perturbation $\dot{T}$ of a circle map T we consider (i) the response of the expectation of an observation function and (ii) the response of isolated spectral points of the transfer operator of T. In each case, under mild conditions on the set of feasible perturbations $\dot{T}$ we show there is a unique optimal feasible infinitesimal perturbation perspective, we remark that even a movemen $\dot{T}_{\mathrm{optimal}}$, maximising the increase of the expectation of the given observation function or maximising the increase of the spectral gap of the transfer operator associated to the system. We derive expressions for the unique maximiser $\dot{T}_{\mathrm{optimal}}$ in terms of its Fourier coefficients. We also devise a Fourier-based computational scheme and apply it to illustrate our theory.

Keywords: linear response, control of statistical properties, transfer operators, spectral properties

Mathematics Subject Classification numbers: 37L30 (primary), 37E10, 37M25, 47A55

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1. Introduction

A C^3 uniformly expanding map of the circle $T : S^1 \to S^1$ is well known to display a linear response to its unique invariant density f_0 . That is, differentiable changes to the map T lead to differentiable changes in f_0 (see [3] for a survey on the subject). Classically, linear response is often phrased as the differentiability of the expectation $\int_{S^1} c(x) \cdot f_0(x)\ \mathrm{d}x$ of an observable $c : S^1 \to \mathbb{R}$. Each infinitesimal perturbation $\dot{T} \in C^3$ of T leads to an infinitesimal perturbation $R(\dot{T})$ of f_0. If $\dot{T}$ is constrained in some meaningful way, for a particular observation c, it is natural to ask whether there is a perturbation $\dot{T}$ that maximises $\int_{S^1} c(x) \cdot R(\dot{T})\ \mathrm{d}x$, and if so, whether such a $\dot{T}$ is unique. For $c \in L^\infty$ we show that there is in fact a unique maximiser under mild conditions on the feasible set of perturbations. When the perturbations $\dot{T}$ are norm-constrained, e.g. by a Sobolev norm $\|\dot{T}\|_{H^4}$ we derive a relatively explicit formula for the unique maximiser $\dot{T}$ in terms of Fourier coefficients.

We pose a similar question for the effect of perturbations on the isolated spectrum of the transfer operator L of T. Perturbations $\dot{T}$ of T lead to perturbations of L, which in turn lead to perturbations of the isolated spectrum and associated eigenprojections. If $\dot{T}$ is constrained in a meaningful way, it is natural to ask whether there is a $\dot{T}$ that maximises the rate of change of the magnitude of an isolated spectral point λ_0. If the isolated spectral point λ_0 is the largest-magnitude spectral point inside the unit circle, it controls the exponential rate of mixing of the system. Therefore one can phrase this spectral optimisation question as 'does there exist a perturbation $\dot{T}$ that maximises the infinitesimal change $\dot{\lambda}$ in the mixing rate?'. In order to answer such quantitative questions, we derive an expression for $\dot{\lambda}$ (the derivative of λ with respect to the perturbation $\dot{T}$) in terms of $\dot{T}$. Under mild conditions on the feasible set of perturbations we show that there is a unique maximiser $\dot{T}$ and when this feasible set is norm constrained in e.g. H^4, we construct an explicit formula for the optimal $\dot{T}$ in terms of its Fourier coefficients.

To numerically estimate the unique perturbation $\dot{T}$ that maximises the expected response of a given observable c, we devise a Fourier-based numerical scheme. This scheme estimates the transfer operator L, the action of the resolvent $(I-L)^{-1}$, and all other derivatives and integrals involved in computing the Fourier coefficients of the unique maximiser $\dot{T}$. To numerically estimate the unique perturbation maximally affecting the mixing rate of the dynamics, we use a related Fourier scheme. In addition to estimating the transfer operator L_0 and its outer spectrum when acting on $W^{1,1}(S^1)$, we also numerically approximate the eigenvector v_0 corresponding to the selected isolated spectral value λ_0, and a representative $\phi_0 \in H^1$ of the corresponding adjoint eigenfunctional $\varphi_0 \in (W^{1,1}(S^1))^*$ of L_0^* acting on $(W^{1,1}(S^1))^*$. Each of the above terms are crucial pieces of the quantitative expression for the objective function we optimise.

Our theory and numerics are illustrated in two examples. In the first example we consider a circle map T with a slightly 'sticky' (derivative near to 1) fixed point at $x = 0$. We show that if the observation c takes large values at $x = 0$, the perturbation $\dot{T}$ retains the fixed point at $x = 0$ and further reduces the derivative, making it more sticky. This increases the proportion of time that orbits spend near $x = 0$ and increases the expectation of the observable. We then show that if the observation function c takes on its maximal value away from $x = 0$, the optimal perturbation $\dot{T}$ sharply moves the fixed point from $x = 0$ in an attempt to weight the invariant density toward larger values of c.

In the second example we construct a circle map with a positive isolated spectral value larger than $1/\inf_x |T'(x)|$. The presence of the relatively large isolated spectral value is due to almost-invariant intervals to the right of $x = 0$ and to the left of $x = 1$. We show that the perturbation $\dot{T}$ that maximally slows the mixing rate (maximises $\dot{\lambda}$) appears to try to strengthen this almost invariance.

The questions posed in this paper are inspired by [1, 2], which considered linear response optimisation for finite-state Markov chains and Hilbert–Schmidt operators, respectively. The latter case includes many random dynamical systems whose associated transfer operator has a kernel, such as deterministic maps with additive noise. In contrast, in this work we consider a class of purely deterministic systems. The study of optimal perturbation and response in this case requires a substantial redevelopment of the optimisation approach, and an entirely distinct and more challenging perturbation theory. Moreover, our approach provides a useful template from which one may treat more complicated deterministic systems where the transfer operator has a spectral gap, such as smooth uniformly hyperbolic maps (in the setting of [19]), including multi-dimensional expanding maps. In particular, a certain amount of technical work was needed to get explicit formulas for the response of the isolated spectrum, see appendix. The question of optimising the outer spectrum (and therefore the mixing rate) has been addressed for flows in the presence of small noise, when the underlying vector field is periodically [14] and aperiodically [13] driven.

Other related works consider the problem of finding the infinitesimal perturbation achieving a prescribed *desired* response direction (in the case of many infinitesimal perturbations achieving the same response one again looks for an optimal one). This problem was also called the 'linear request problem' and studied in [16, 17, 23] from a theoretical point for some classes of deterministic and random maps, and in applications in cellular automata [26] and climate [7]. In particular, the work [17] considers the case of expanding maps and the problem of finding a minimum-norm infinitesimal perturbation resulting in a given response for the invariant density of the system.

An outline of the paper is as follows. In section 2 we formalise the class of dynamical systems and perturbations we consider. Proposition 3 summarises the fundamental theory concerning differentiability of invariant densities and spectral points for expanding maps. Section 3 recaps relevant results from convex optimisation. Section 4 formally sets up the optimisation problem to maximise the rate of change of the expectation of an observation function c. Proposition 7 states that there is a unique maximiser $\dot{T}$ and theorem 8 provides explicit expressions for the Fourier coefficients of the optimal $\dot{T}$. Section 5 sets up the response maximisation problem for isolated spectral values λ. Proposition 9 verifies there is a unique optimal perturbation $\dot{T}$ and theorem 11 states explicit formulae for the Fourier coefficients of the optimal $\dot{T}$ and the optimal value of the corresponding response of λ. Section 6 describes our computational approach to estimate all of the relevant objects required to numerically solve the two main optimisation problems, and section 7 illustrates our theory and numerics through two examples. Finally, in the appendix, we recall some known results about linear response of invariant measures and resolvent operators, and from these facts we derive general response formulas for invariant measures and isolated eigenvalues we use in the paper.

2. Linear response of invariant densities and isolated eigenvalues

In this section we consider the response of the physical measure of an expanding circle map, and of the leading eigenvalues of the associated transfer operator, to deterministic perturbations of the map. We begin by setting up the class of dynamical systems we will consider. Proposition 2 then recalls known continuity properties for the physical measure and isolated spectrum under deterministic perturbations of the map, and proposition 3 states the corresponding linear response results. proposition 3 also contains explicit formulas for the response of these objects under suitable deterministic perturbations of the system. These explicit formulas will be used in the computation of the optimal (response-maximising) perturbation.

Several linear response results in the literature treat perturbations that compose the dynamics with a diffeomorphism near to the identity, e.g. [3]. In this paper we consider additive perturbations applied directly to the map, as we believe this is natural for applications.

Let us consider some $\bar{\delta} > 0$ and a family of C^3 maps $\{T_\delta : S^1 \to S^1\}_{\delta \in [0,\bar{\delta})}$ satisfying the following assumptions:

(A0) $T_0 \in C^4(S^1, S^1)$.
(A1) There is $0 < \alpha < 1$ such that $T_0'(x) \geqslant \alpha^{-1}$ for all $x \in S^1$[3].
(A2) The dependence of the family T_δ on δ is differentiable at $\delta = 0$ in the following strong sense:

$$T_\delta(x) = T_0(x) + \delta \dot{T}(x) + O_{C^3}\left(\delta^2\right) \tag{1}$$

where $\dot{T} \in C^3(S^1, \mathbb{R})$ and we say a family of functions $F_\delta \in C^3(S^1, \mathbb{R})$ is $O_{C^3}(\delta^2)$ if

$$\sup_{\delta \in \left(0, \bar{\delta}\right)} \frac{||F_\delta||_{C^3}}{\delta^2} < \infty. \tag{2}$$

We study the statistical properties of these map perturbations through their associated transfer operators. It is well known that if we consider the action of L_δ on a suitable Sobolev space, this operator is quasi-compact (see e.g. [25]). We denote the Sobolev space of functions having weak kth derivatives in L^p by $W^{k,p}$ (see [12] or [9] for the general theory of such spaces). We recall the definition of the transfer operator associated to a map of the circle and of the derivative operator associated to perturbations as in (A2).

Definition 1. The transfer operator $L_\delta : W^{1,1}(S^1, \mathbb{C}) \to W^{1,1}(S^1, \mathbb{C})$ associated to an expanding map as above $T_\delta : S^1 \to S^1$ is defined by

$$L_\delta f(x) = \sum_{y \in T_\delta^{-1} x} \frac{f(y)}{T_\delta'(y)}$$

for each $f \in W^{1,1}(S^1, \mathbb{C})$. The derivative operator $\dot{L} : W^{1,1}(S^1, \mathbb{C}) \to L^1(S^1, \mathbb{C})$ associated to a map perturbation in the direction $\dot{T} \in C^3(S^1, \mathbb{R})$ is defined as

$$\dot{L}(f) := -L_0\left(\left[\frac{f\dot{T}}{T_0'}\right]'\right). \tag{3}$$

For the moment, we simply call $\dot{L}$ the 'derivative operator'; in appendix A.4.3 we show that $\dot{L}$ indeed arises as a derivative of the family of operators L_δ at $\delta = 0$. We denote by f_δ an invariant probability density for T_δ. Such densities are fixed points of the operators L_δ; that is, $L_\delta f_\delta = f_\delta$. It is well known that an expanding map has a unique invariant probability density which is absolutely continuous with respect to the Lebesgue measure on S^1 (see e.g. [24, 25]).

We now recall a series of facts about stability (proposition 2) and linear response (proposition 3) of the invariant density and isolated spectrum of expanding maps when subjected

[3] T_0 is a map between one-dimensional manifolds. Its differential at a point $x \in S^1$ is a linear map between the tangent spaces at x and $T_0(x)$, hence a linear map : $\mathbb{R} \to \mathbb{R}$. Its slope will be denoted by $T_0'(x) \in \mathbb{R}$. The same notation will be applied when considering maps $f : S^1 \to \mathbb{R}$ and their derivatives.

to deterministic perturbations. The following proposition collects well-known facts about the spectral picture of expanding maps and its stability[4].

Proposition 2. *Consider a family of maps $T_\delta : S^1 \to S^1$ for $\delta \in [0,\overline{\delta})$ satisfying (A0), (A1) and (A2). Let $L_\delta : W^{1,1}(S^1,\mathbb{C}) \to W^{1,1}(S^1,\mathbb{C})$ be the transfer operators associated to T_δ.*

(I) *There is a unique invariant probability density $f_\delta \in W^{2,1}(S^1,\mathbb{R})$ of T_δ; furthermore $||f_\delta - f_0||_{W^{1,1}} \to 0$ as $\delta \to 0$.*
(II) *If $\lambda_0 \in \mathbb{C}$ is an eigenvalue of L_0 acting on $W^{1,1}(S^1,\mathbb{C})$ such that $\alpha < |\lambda_0| < 1$, then λ_0 is isolated. Furthermore, if λ_0 is simple, there is $\delta_* > 0$ such that for $\delta \in [0,\delta_*)$, L_δ has a simple isolated eigenvalue λ_δ and the map $\delta \mapsto \lambda_\delta$ is continuous.*

It is well known that the speed of exponential mixing of an expanding map is quantitatively related to the spectral gap of L_0. Sometimes the spectral gap is due to the existence of an isolated eigenvalue λ_0 satisfying $\alpha < |\lambda_0| < 1$. When an isolated eigenvalue exists (also called a Pollicott–Ruelle resonance) with magnitude close to 1, the signed distribution of the corresponding eigenfunction can be viewed as encoding an obstruction to exponential mixing at a rate suggested by the rate of local expansion in phase space; see [11] for an elaboration of this idea. This motivates our study of the linear response of eigenvalues that control the exponential rate of mixing. Our optimal linear response theory constructs optimal deterministic perturbations of the map, which reduce the magnitude of $|\lambda_0|$ at as rapidly as possible.

Linear response of the invariant measure of expanding maps under deterministic perturbations is a classical result (see [3]). A response formula for deterministic additive perturbations (see (1)) similar to (4) was given in [17]. Differentiability of the isolated eigenvalues of the transfer operator associated to expanding maps under deterministic perturbations is due to [19] (see also [4]). Since we need an explicit formula for this derivative, in proposition 3 we also provide such a formula (see (5)). To the best of our knowledge, the formula (5) is new in the deterministic setting. It mirrors the expression in corollary III.11 [20], which in our notation applies to a family of quasi-compact operators L_δ that are continuously differentiable with respect to δ in operator norm. The operator perturbations induced by a differentiable family T_δ of maps satisfying (A0)–(A2) do not fit into this framework and require a more careful treatment, this will be done using theory developed in [20] and [19], as laid out in the appendix.

Proposition 3. *Consider a family of maps $T_\delta : S^1 \to S^1$ for $\delta \in (0,\overline{\delta})$ satisfying (A0)–(A2) with associated invariant densities f_δ as above.*

(I) The following linear response formula for the invariant measure f_δ holds:

$$\lim_{\delta\to 0}\left\|\frac{f_\delta - f_0}{\delta} + (Id - L_0)^{-1} L_0\left(\left[\frac{f_0\dot{T}}{T_0'}\right]'\right)\right\|_{L^1} = 0. \tag{4}$$

(II) Let $\lambda_0 \in \mathbb{C}$ be a simple, isolated eigenvalue of L_0 acting on $W^{1,1}(S^1,\mathbb{C})$ such that $\alpha < |\lambda_0| < 1$. Let $\varphi_0 \in (W^{1,1}(S^1,\mathbb{C}))^*$ be the eigenfunction of the adjoint[5] operator L_0^* :

[4] See [25] or [15] for details on expanding maps, the invariant measures and its regularity, their stability under deterministic perturbations and their spectral picture. The spectral stability the other results stated at item (II) can be obtained applying to such systems the classical spectral stability results of [21].

[5] Given $L_0 : W^{1,1}(S^1,\mathbb{C}) \to W^{1,1}(S^1,\mathbb{C})$. If we denote by $(W^{1,1}(S^1,\mathbb{C}))^*$ the dual of $W^{1,1}(S^1,\mathbb{C})$ the adjoint operator is a linear operator $L_0^* : (W^{1,1}(S^1,\mathbb{C}))^* \to (W^{1,1}(S^1,\mathbb{C}))^*$ defined in the following way: for $\omega_1 \in (W^{1,1}(S^1,\mathbb{C}))^*$ we have $L_0^*(\omega_1) = \omega_2$ where for each $f \in W^{1,1}(S^1,\mathbb{C})$ one has $\omega_2(f) = \omega_1(L_0(f))$.

$(W^{1,1}(S^1,\mathbb{C}))^* \to (W^{1,1}(S^1,\mathbb{C}))^*$ associated to λ_0 i.e. $L_0^*\varphi_0 = \lambda_0\varphi_0$, where φ_0 is scaled so that $\varphi_0(\mathbf{1}) = 1$. Let $v_0 \in W^{1,1}(S^1,\mathbb{C})$ be the eigenfunction of L_0 associated to λ_0 scaled so that $\varphi_0(v_0) = 1$ (in fact $v_0 \in W^{3,1}(S^1,\mathbb{C})$). Furthermore, consider the continuous map $\delta \mapsto \lambda_\delta$ as in (II) of proposition 2. This map is differentiable at 0 and

$$\frac{\mathrm{d}\lambda_\delta}{\mathrm{d}\delta}|_{\delta=0} = \varphi_0\left(\dot{L}(v_0)\right). \tag{5}$$

The proof of proposition 3 is postponed to appendix A.5.

3. Recap of convex optimisation

We will consider a set $P \subset C^3(S^1,\mathbb{R})$ of allowed infinitesimal perturbations $\dot{T}$ to the map T_0. We are interested in selecting an *optimal* perturbation $\dot{T}$ in terms of (i) maximising the rate of change of the expectation of a chosen observable, and (ii) maximising the rate of change in the magnitude of an isolated eigenvalue. Because we wish to perform an optimisation we consider P inside some Hilbert space $\mathcal{H}$. We will also assume that P is bounded and convex, which we believe are natural hypotheses; convexity because if two different perturbations of a system are possible, then their convex combination – applying the two perturbations with different intensities—should also be possible.

Definition 4. We say that a convex closed set $P \subseteq \mathcal{H}$ is *strictly convex* if for each pair $x, y \in P$ and for all $0 < \gamma < 1$, the points $\gamma x + (1-\gamma)y \in \mathrm{int}(P)$, where the relative interior[6] is meant.

We briefly recall some relevant results from convex optimisation. Suppose $\mathcal{H}$ is a separable Hilbert space and $P \subset \mathcal{H}$. Let $\mathcal{J} : \mathcal{H} \to \mathbb{R}$ be a continuous linear function. Consider the abstract problem to find $p_* \in P$ such that

$$\mathcal{J}(p_*) = \max_{p\in P} \mathcal{J}(p). \tag{6}$$

The existence and uniqueness of an optimal perturbation follows from properties of P.

Proposition 5 (existence of the optimal solution). *Let P be bounded, convex, and closed in $\mathcal{H}$. Then problem (6) has at least one solution.*

Upgrading convexity of the feasible set P to strict convexity provides uniqueness of the optimum.

Proposition 6 (Uniqueness of the optimal solution). *Suppose P is closed, bounded, and strictly convex subset of $\mathcal{H}$, and that P contains the zero vector in its relative interior. If $\mathcal{J}$ is not uniformly vanishing on P then the optimal solution to (6) is unique.*

Note that in the case $\mathcal{J}$ is uniformly vanishing, all the elements of P are solutions of the problem (6). See lemma 6.2 [13], proposition 4.3 [2], or corollary 3.23 of [9] for the proofs of propositions 5 and 6 and more details.

4. Optimally increasing the expectation of an observation

Let $c \in L^\infty(S^1,\mathbb{R})$ be an observable. We consider the problem of finding an infinitesimal perturbation $\dot{T}$ of our map T_0 that maximises the rate of change of expectation of c. If c were an

[6] The relative interior of a closed convex set C is the interior of C relative to the closed affine hull of C, see e.g. [8].

indicator, for example, one could control the invariant density toward the support of c. Given a family of maps T_δ satisfying (A0)–(A2) with invariant densities f_δ, we denote the response of the system to $\dot{T}$ by

$$R\left(\dot{T}\right) := \lim_{\delta\to 0}\frac{f_\delta - f_0}{\delta}.$$

This limit is converging in L^1 as proved in proposition 3. Under our assumptions we easily get

$$\lim_{\delta\to 0}\frac{\int c\left(x\right)f_\delta\left(x\right)\,\mathrm{d}x - \int c\left(x\right)f_0\left(x\right)\,\mathrm{d}x}{\delta} = \int c\left(x\right)R\left(\dot{T}\right)\left(x\right)\,\mathrm{d}x. \tag{7}$$

Hence the rate of change of the expectation of c with respect to δ is given by the linear response of the system under the given perturbation. To take advantage of the general results of the previous section, we perform the optimisation of $\dot{T}$ over a closed, bounded, convex subset of a suitable Hilbert space $\mathcal{H}$ containing the zero perturbation. Because we require $\dot{T}\in C^3$ in proposition 3 we select $\mathcal{H} = H^4(S^1,\mathbb{R})$[7] and consider a convex closed set $P\subseteq\mathcal{H}$. To maximise the RHS of (7) we set $\mathcal{J}(\dot{T}) := \int c(x)\cdot R(\dot{T})(x)\,\mathrm{d}x$ and set P to be the unit ball in $H^4(S^1,\mathbb{R})$, which is bounded, convex, and contains the zero vector. We hence consider the problem

$$\max_{\dot{T}\in H^4}\quad \mathcal{J}\left(\dot{T}\right) \tag{8}$$

$$\text{subject to}\quad \|\dot{T}\|_{H^4}\leqslant 1. \tag{9}$$

We use the following notation to indicate the zero-average spaces in $W^{i,1}$. For $i\in\{0,1,2\}$

$$V_i := \left\{g\in W^{i,1}\left(S^1\right)\ s.t.\ \int_{S^1} g\,\mathrm{d}m = 0\right\}.$$

Proposition 7. *If $\mathcal{J}$ is not uniformly vanishing, there is a unique optimum map perturbation $\dot{T}$ for Problem (8).*

Proof. The result will directly follow from propositions 5 and 6, once we verify that $\mathcal{J} : H^4\to\mathbb{R}$ is continuous. Since

$$R\left(\dot{T}\right) = -\left(I-L_0\right)^{-1}\left(L_0\left[\left(\frac{f_0\dot{T}}{T_0'}\right)'\right]\right)$$

we have $||R(\dot{T})||_{L^1}\leqslant||R(\dot{T})||_{W^{1,1}}\leqslant||(I-L_0)^{-1}||_{V_1\to W^{1,1}}\cdot\|L_0\|_{W^{1,1}\to W^{1,1}}\cdot||\left(\frac{f_0\dot{T}}{T_0'}\right)'||_{W^{1,1}}$. The first two terms in the product are well known to be bounded in our case (see proposition 19 and lemma 18). Because $f_0\in W^{3,1}$ and $T_0'\in C^3$ is uniformly bounded below, there exists C_2 such that $\left\|\left(\frac{f_0\dot{T}}{T_0'}\right)'\right\|_{W^{1,1}}\leqslant C_2||\dot{T}||_{H^4}$ for some $C_2\geqslant 0$. Thus, by linearity of $R(\dot{T})$, one sees that $\dot{T}\mapsto\int c(x)\cdot R(\dot{T})(x)\,\mathrm{d}x$ is continuous. □

[7] Recall that $H^4\subseteq C^3$ because we are in dimension 1; see e.g. [9].

4.1. An explicit formula for the optimal perturbation

We now state a theorem identifying this optimal perturbation for the problem (8).

Theorem 8. *Let $c \in L^\infty(S^1, \mathbb{R})$ be an observation function and the family of maps T_δ satisfy (A0), (A1) and (A2). The perturbation $\dot{T} \in H^4(S^1, \mathbb{R})$ that maximises the expected linear response $\int_{S^1} c(x) R(\dot{T})(x)\, \mathrm{d}x$ (solves problem (8) and (9)) is given by $\dot{T} = \sum_{n=-\infty}^{\infty} a_n e_n$, where $e_n(x) = \exp(2\pi i n x)$ and the coefficients a_n are given by*

$$a_n = \left[- \int c \cdot \overline{(I - L_0)^{-1} L_0 \left(\left[\frac{e_n f_0}{T'} \right]' \right)} \right] \Bigg/ 2\nu \left(\sum_{k=0}^{4} \left(4\pi^2 n^2\right)^k \right), \tag{10}$$

where $\nu > 0$ is chosen so that $\|\dot{T}\|_{H^4} = 1$; that is $\sum_{n=-\infty}^{\infty} \sum_{k=0}^{4} (4\pi^2 n^2)^k a_n^2 = 1$.

Proof. We write the continuous (by the proof of proposition 7) linear functional $\mathcal{J} : H^4 \to \mathbb{R}$ as an L^2 inner product

$$\mathcal{J}\left(\dot{T}\right) = \left\langle c, -\left(I - L_0\right)^{-1} L_0 \left(\left(\frac{f_0 \dot{T}}{T_0'} \right)' \right) \right\rangle,$$

Define a second continuous linear functional $g : H^4 \to \mathbb{R}$ using the norm constraint: $g(\dot{T}) = \|\dot{T}\|_{H^4}^2 - 1$.

For a general continuous linear functional $\mathcal{F} : H^4 \to \mathbb{R}$ we denote by $\Delta\mathcal{F}(\dot{T}, \tilde{T})$ the Gâteaux variation of the functional $\mathcal{F}$ at $\dot{T}$ in the direction $\tilde{T}$, i.e. $\Delta\mathcal{F}(\dot{T}, \tilde{T}) = \lim_{h \to 0} \mathcal{F}(\dot{T} + h\tilde{T})/h$. Given $\dot{T} \in H^4$, $\Delta\mathcal{J}(\dot{T}, \tilde{T})$ exists for all $\tilde{T} \in H^4$ by linearity of $\mathcal{J}$; indeed

$$\Delta\mathcal{J}\left(\dot{T}, \tilde{T}\right) = \mathcal{J}\left(\tilde{T}\right) = \left\langle c, -\left(I - L_0\right)^{-1} L_0 \left(\left(\frac{f_0 \tilde{T}}{T_0'} \right)' \right) \right\rangle \qquad \text{for all } \dot{T}, \tilde{T} \in H^4. \tag{11}$$

The variation of the functional g is defined similarly, and it is straightforward to show that

$$\Delta g\left(\dot{T}, \tilde{T}\right) = 2\langle \dot{T}, \tilde{T} \rangle_{H^4} \qquad \text{for all } \dot{T}, \tilde{T} \in H^4. \tag{12}$$

A variation $\Delta\mathcal{F}$ is called *weakly continuous* if $\lim_{\dot{S} \to \dot{T}} \Delta\mathcal{F}(\dot{S}, \tilde{T}) = \Delta\mathcal{F}(\dot{T}, \tilde{T})$ for each $\tilde{T}$. It is clear from the explicit expressions given above that both $\Delta\mathcal{J}$ and Δg are weakly continuous variations.

We form the Lagrangian

$$\mathcal{L}\left(\dot{T}, \nu\right) = \mathcal{J}\left(\dot{T}\right) - \nu g\left(\dot{T}\right), \tag{13}$$

with Lagrange multiplier $\nu \in \mathbb{R}$. Having established weak continuity of the variations of $\mathcal{J}$ and g, the Euler–Lagrange Multiplier Theorem [27] (section 3.3), guarantees that a necessary condition for $\dot{T}$ be a local extremum of the constrained problem (8) is that:

$$\Delta\mathcal{L}\left(\dot{T}, \tilde{T}\right) = 0 \qquad \text{for all } \tilde{T} \in H^4. \tag{14}$$

and

$$g\left(\dot{T}\right) = 0.$$

We express $\dot{T} \in H^4$ as

$$\dot{T} = \sum_{m=-\infty}^{\infty} a_m e_m,$$

with $a_m \in \mathbb{C}$.

By continuity of $\Delta\mathcal{J}(\dot{T},\cdot) : H^4 \to \mathbb{R}$ and density of smooth functions in H^4, we may equivalently insist that (14) holds for each $\tilde{T} = e_n = \exp(2\pi i n x)$. That is, for each $n \in \mathbb{Z}$ one has (using (11)–(14)):

$$\begin{aligned}
0 &= \int c \cdot \overline{-(I-L_0)^{-1} L_0 \left(\left[\frac{e_n f_0}{T_0'} \right]' \right)} - 2\nu \sum_{k=0}^{4} \int \left(\sum_m a_m e_m \right)^{(k)} \cdot \overline{(e_n)^{(k)}} \\
&= \int c \cdot \overline{-(I-L_0)^{-1} L_0 \left(\left[\frac{e_n f_0}{T_0'} \right]' \right)} - 2\nu \sum_{k=0}^{4} \int \left(\sum_m (2\pi i m)^k a_m e_m \right) \cdot (-2\pi i n)^k \overline{(e_n)} \\
&= \int c \cdot \overline{-(I-L_0)^{-1} L_0 \left(\left[\frac{e_n f_0}{T_0'} \right]' \right)} - 2\nu \left(\sum_{k=0}^{4} \left(4\pi^2 n^2\right)^k \right) a_n,
\end{aligned}$$

where final equality uses $\int e_m \cdot \overline{e_n} = \delta_{m,n}$. Thus, for $n \in \mathbb{Z}$, the coefficients a_n are given by

$$a_n = \left[-\int c \cdot \overline{(I-L_0)^{-1} L_0 \left(\underbrace{\left[\frac{e_n f_0}{T_0'} \right]'}_{=:b_n} \right)} \right] \Big/ 2\nu \left(\sum_{k=0}^{4} \left(4\pi^2 n^2\right)^k \right), \tag{15}$$

where ν is chosen so that $\|\dot{T}\|_{H^4} = 1$ to satisfy $g(\dot{T}) = 0$; that is $\sum_{n=-\infty}^{\infty} \sum_{k=0}^{4} (4\pi^2 n^2)^k a_n^2 = 1$. Notice that each b_n has zero mean as the derivative of any periodic function integrates to zero. Further note that the integral in (15) makes sense, because $e_n \in C^\infty$, $f_0 \in C^3$ (since $T_0 \in C^4$) and $T_0' \in C^3$, and therefore the function b_n is in C^2 (in particular uniformly bounded with $|b_n|_\infty = O(n)$ due to the derivative acting on e_n) and in $W^{1,1}$.

We now verify that the above a_n define a $\dot{T} \in H^4$. Because we divide by n^8 in (15), the above growth estimate of b_n leads to a decay rate of $|a_n| \leqslant C/n^7$. It is a fact that if the Fourier coefficients $a_n(f)$ of some function f satisfy $a_n(f) = O(1/n^{k+1+\gamma})$ for $\gamma > 0$ then $f \in C^k$. Thus the optimal perturbations $\dot{T}$ with Fourier coefficients given by a_n in (15) lie in $C^5 \subset H^4$.

Finally, we determine the sign of the Lagrange multiplier ν. Because of the form of (13) and the fact that we maximise (rather than minimise) $\mathcal{J}$, it will turn out that $\nu > 0$. Setting $\tilde{T} = \dot{T}$ in (14) and using (11) and (12) we have

$$0 = \Delta\mathcal{J}\left(\dot{T},\dot{T}\right) - \nu \cdot \Delta g\left(\dot{T},\dot{T}\right) = \mathcal{J}\left(\dot{T}\right) - 2\nu \cdot \langle \dot{T},\dot{T} \rangle_{H^4}.$$

Because $\mathcal{J}(\dot{T})$ is maximal, we have $\mathcal{J}(\dot{T}) > 0$. The above display equation then implies that $\nu > 0$. □

5. Optimally moving isolated spectral points

We again consider a family of maps $\{T_\delta\}$ satisfying (A0), (A1) and (A2), and assume that L_0 has a simple eigenvalue λ_0 satisfying $|\lambda_0| > \alpha$. By proposition 3 the corresponding eigenfunction v_0 lies in $W^{3,1}$. Proposition 2 guarantees the existence of a continuous family of simple eigenvalues λ_δ for the perturbed operators $L_\delta : W^{1,1} \to W^{1,1}$. Proposition 3 then show the differentiability of this family, providing a formula for its derivative at $\delta = 0$, namely $\dot{\lambda}(\dot{T}) = \varphi_0(\dot{L}(\dot{T})v_0)$.

We wish to optimise $\dot{\lambda}(\dot{T})$ as a function of the map perturbation $\dot{T}$. This optimisation will be performed on the separable Hilbert space $H^4(S^1) \subset C^3(S^1)$. We therefore define a linear functional $\mathcal{J} : H^4(S^1, \mathbb{R}) \to \mathbb{C}$ as

$$\mathcal{J}\left(\dot{T}\right) := \dot{\lambda}\left(\dot{T}\right) = \varphi_0\left(\dot{L}\left(\dot{T}\right)v_0\right). \tag{16}$$

To simplify the explicit formulae appearing later in section 5.1 for the optimal perturbation $\dot{T}$, we assume that λ_0 is real and positive. We therefore wish to select a perturbation $\dot{T}$ of T_0 so as to maximise the rate of increase of λ_0 under the perturbation $\dot{T}$ (in other words, maximise $\dot{\lambda}$):

$$\max_{\dot{T} \in H^4} \quad \mathcal{J}\left(\dot{T}\right) \tag{17}$$

$$\text{subject to} \quad \|\dot{T}\|_{H^4} \leqslant 1. \tag{18}$$

Proposition 9. *If $\mathcal{J}$ is not uniformly vanishing there is a unique optimal map perturbation $\dot{T}$ for Problem (17) and (18).*

Proof. Because $H^4(S^1)$ is a separable Hilbert space and the unit ball in $H^4(S^1)$ is a strictly convex, bounded, closed set containing the zero element, in order to apply propositions 5 and 6, it is sufficient to check that $\mathcal{J} : H^4 \to \mathbb{R}$ is continuous. We have to verify that for fixed $\varphi_0 \in (W^{1,1})^*$ and $v_0 \in W^{3,1}$ (see proposition 3), there is a $C < \infty$ such that $|\mathcal{J}(\dot{T})| \leqslant C\|\dot{T}\|_{H^4}$. We have

$$|\mathcal{J}\left(\dot{T}\right)| = |\varphi_0\left(\dot{L}\left(\dot{T}\right)v_0\right)| \leqslant \|\varphi_0\|_{(W^{1,1})^*}\|\dot{L}\left(\dot{T}\right)v_0\|_{W^{1,1}},$$

and so it is sufficient to show that $\|\dot{L}(\dot{T})v_0\|_{W^{1,1}} \leqslant C\|\dot{T}\|_{H^4}$. We note that since $T_0 \in C^4$ with $T_0' > 1$ and $v_0 \in W^{3,1}$ we have $v_0/T_0' \in W^{3,1}$. Now

$$\begin{aligned}\|\dot{L}\left(\dot{T}\right)v_0\|_{W^{1,1}} &= \|L_0\left(\left(\dot{T}v_0/T_0'\right)'\right)\|_{W^{1,1}} \\ &\leqslant C\|\left(\dot{T}v_0/T_0'\right)'\|_{W^{1,1}} \quad \text{by lemma 18.}\end{aligned}$$

Thus we can conclude that there is a constant $\tilde{C} > 0$ such that

$$\|\dot{L}\left(\dot{T}\right)v_0\|_{W^{1,1}} \leqslant \tilde{C}\|\dot{T}\|_{H^4}.$$

□

5.1. Explicit formula for optimal solution

We wish to maximise $\mathcal{J} : H^4(S^1, \mathbb{R}) \to \mathbb{R}$, where $\mathcal{J}(\dot{T}) = \varphi_0(\dot{L}(\dot{T})v_0)$, subject to $\|\dot{T}\|_{H^4} = 1$. From proposition 9 we know there is a unique optimum. In order to write an explicit formula for the optimal $\dot{T}$ we require a representative of $\varphi_0 \in (W^{1,1}(S^1, \mathbb{R}))^*$ when acting on elements of $H^1(S^1, \mathbb{R})$.

Lemma 10. *Let $\varphi_0 \in (W^{1,1}(S^1, \mathbb{R}))^*$. Then there is a $\phi_0 \in H^1(S^1, \mathbb{R})$ such that*

$$\varphi_0(f) = \int_{S^1} \phi_0 \bar{f}\,\mathrm{d}x + \int_{S^1} \phi_0' \bar{f}'\,\mathrm{d}x \quad \textit{for all } f \in H^1\left(S^1, \mathbb{R}\right). \tag{19}$$

Proof. Since $H^1(S^1) \subseteq W^{1,1}(S^1)$ we have that $(W^{1,1}(S^1))^* \subseteq (H^1(S^1))^*$ and therefore $\varphi_0 \in (H^1(S^1))^*$. The result follows by Riesz representation theorem. □

We now state a theorem identifying this optimal perturbation.

Theorem 11. *Let the family of maps T_δ satisfy (A0), (A1) and (A2) and consider a family of isolated eigenvalues λ_δ. The perturbation $\dot{T} \in H^4(S^1,\mathbb{R})$ that maximises the expected linear response $\dot{\lambda}(\dot{T})$ of λ_0 (i.e. solves problem (17) and (18)) is given by $\dot{T} = \sum_{n=-\infty}^{\infty} a_n e_n$, where $e_n(x) = \exp(2\pi i n x)$ and the coefficients a_n are given by*

$$a_n = \left[-\int \phi_0 \cdot \overline{L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)} - \int \phi_0' \cdot \overline{\left(L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)\right)'}\right] \Big/ 2\nu\left(\sum_{k=0}^{4}\left(4\pi^2 n^2\right)^k\right), \tag{20}$$

where $\nu > 0$ is chosen so that $\|\dot{T}\|_{H^4} = 1$; that is $\sum_{n=-\infty}^{\infty}\sum_{k=0}^{4}(4\pi^2n^2)^k a_n^2 = 1$.

Moreover, the maximal linear response is given by

$$\dot{\lambda}\left(\dot{T}\right) = -\int \phi_0 \cdot \overline{L_0\left(\left(\frac{\dot{T}v_0}{T'}\right)'\right)} - \int \phi_0' \cdot \overline{\left(L_0\left(\left(\frac{\dot{T}v_0}{T'}\right)'\right)\right)'}.$$

Proof. We follow a similar strategy to the proof of theorem 8. Given $\dot{T} \in H^4$, we first need to show that $\Delta\mathcal{J}(\dot{T},\tilde{T}) = \lim_{h\to 0}(\mathcal{J}(\dot{T}+h\tilde{T}) - \mathcal{J}(\dot{T}))/h$ exists for each $\tilde{T} \in H^4$. Since $\tilde{T} \in C^3$ and $v_0 \in C^3$, proposition 22 yields $\dot{L}(\tilde{T})v_0 \in H^1$. Thus by lemma 10, and recalling (16), there is a $\phi_0 \in H^1(S^1,\mathbb{R})$ such that

$$\mathcal{J}\left(\tilde{T}\right) = \varphi_0\left(\dot{L}\left(\tilde{T}\right)v_0\right) = \int \phi_0 \cdot \overline{\dot{L}\left(\tilde{T}\right)v_0} + \phi_0' \cdot \overline{\left(\dot{L}\left(\tilde{T}\right)v_0\right)'},$$

which is finite for each $\tilde{T} \in H^4$, and so by linearity of $\mathcal{J}$, we see that for each $\dot{T},\tilde{T} \in H^4$,

$$\begin{aligned}\Delta\mathcal{J}\left(\dot{T},\tilde{T}\right) = \phi_0\left(\dot{L}\left(\tilde{T}\right)v_0\right) &= \int \phi_0 \cdot \overline{\dot{L}\left(\tilde{T}\right)v_0} + \phi_0' \cdot \overline{\left(\dot{L}\left(\tilde{T}\right)v_0\right)'} \\ &= -\int \phi_0 \cdot \overline{L_0\left(\left(\frac{\tilde{T}v_0}{T'}\right)'\right)} - \phi_0' \cdot \overline{\left[L_0\left(\left(\frac{\tilde{T}v_0}{T'}\right)'\right)\right]'}.\end{aligned}$$

The variation of the functional g is handled identically to the proof of theorem 8; one obtains

$$\Delta g\left(\dot{T},\tilde{T}\right) = 2\langle\dot{T},\tilde{T}\rangle_{H^4} \qquad \text{for all } \dot{T},\tilde{T} \in H^4.$$

Weak continuity of the variations $\Delta\mathcal{J}$ and Δg follows as in the proof of theorem 8.

We form the Lagrangian

$$\mathcal{L}\left(\dot{T},\nu\right) = \mathcal{J}\left(\dot{T}\right) - \nu g\left(\dot{T}\right),$$

with Lagrange multiplier $\nu \in \mathbb{R}$. Having established weak continuity of the variations of $\mathcal{J}$ and g, the Euler–Lagrange multiplier theorem [27] (section 3.3), guarantees that a necessary condition for $\dot{T}$ be a local extremum of the constrained problem (17) and (18) is that:

$$\Delta\mathcal{L}\left(\dot{T},\tilde{T}\right) = 0 \qquad \text{for all } \tilde{T} \in H^4. \tag{21}$$

and

$$g\left(\dot{T}\right) = 0.$$

We express $\dot{T} = \sum_{m=-\infty}^{\infty} a_m e_m$, with $a_m \in \mathbb{C}$. By continuity of $\Delta\mathcal{J}(\dot{T},\cdot) : H^4 \to \mathbb{C}$, and density of smooth functions in H^4 we may equivalently insist that (21) holds for each $\tilde{T} = e_n = \exp(2\pi i n x)$. That is, for each $n \in \mathbb{Z}$ one has (using (21)):

$$\begin{aligned}
0 &= -\left[\int \phi_0 \cdot \overline{L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)}\right] - \left[\int \phi_0' \cdot \overline{\left(L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)\right)'}\right] \\
&\quad - 2\nu \sum_{k=0}^{4} \left[\int \left(\sum_m a_m e_m\right)^{(k)} \cdot \overline{(e_n)^{(k)}}\right] \\
&= -\int \phi_0 \cdot \overline{L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)} - \int \phi_0' \cdot \overline{\left(L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)\right)'} \\
&\quad - 2\nu \sum_{k=0}^{4} \int \left(\sum_m (2\pi i m)^k a_m e_m\right) \cdot (-2\pi i n)^k \overline{(e_n)} \\
&= -\int \phi_0 \cdot \overline{L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)} - \int \phi_0' \cdot \overline{\left(L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)\right)'} - 2\nu\left(\sum_{k=0}^{4} \left(4\pi^2 n^2\right)^k\right) a_n,
\end{aligned}$$

where final equality uses $\int e_m \cdot \overline{e_n} = \delta_{m,n}$. Thus, for $n \in \mathbb{Z}$, the coefficients a_n are given by

$$a_n = \left[-\int \phi_0 \cdot \overline{L_0\left(\left(\frac{e_n v_0}{T'}\right)'\right)} - \int \phi_0' \cdot \overline{\left(L_0\left(\underbrace{\left(\frac{e_n v_0}{T'}\right)'}_{=:b_n}\right)\right)'}\right] \Bigg/ 2\nu\left(\sum_{k=0}^{4} \left(4\pi^2 n^2\right)^k\right), \tag{22}$$

where ν is chosen so that $\|\dot{T}\|_{H^4} = 1$; that is $\sum_{n=-\infty}^{\infty} \sum_{k=0}^{4} (4\pi^2 n^2)^k a_n^2 = 1$. Notice that each b_n has zero mean as the derivative of any periodic function integrates to zero. The integrals in (22) makes sense, because $e_n \in C^\infty$, $v_0 \in C^3$ and $T' \in C^3$ and is bounded uniformly below by 1. Therefore the function b_n is in C^1 (in particular uniformly bounded with $|b|_\infty = O(n)$ due to the derivative acting on e_n) and in H^1. We verify that the above a_n define a $\dot{T} \in H^4$ in exactly the same way as in theorem 8. The fact that $\nu > 0$ follows exactly as at the end of the proof of theorem 8. The final claim of the theorem follows from (16), using lemma 10 to represent $\varphi_0(\dot{L}(\dot{T})v_0)$. □

6. Numerical approach

The computations revolve around evaluating the integrals in (10) and (20). We begin by discussing the common elements of these computations and then discuss specific elements in the subsequent subsections. In this numerical section we will denote L_0 by simply L to avoid confusion with the approximate transfer operator at resolution N, which we denote by L_N.

We first build a projection of $\hat{L}$ (the action of L in frequency space) on complex exponentials $e_n(x) = \exp(2\pi i n x)$. A finite spatial grid $\{0, 1/N, 2/N, \ldots, (N-1)/N\} \subset S^1$ corresponds to the N Fourier modes $\{e_{-N/2+1},\ldots,e_{N/2}\}$ via discrete Fourier transform. A matrix $\hat{L}_N$ is constructed from estimates of the integrals

$$\hat{L}_{N,nm} := \int_{S^1} L(e_m)\cdot\bar{e}_n = \int_{S_1} e_m\cdot\bar{e}_n\circ T = \overline{\int_{S_1}\bar{e}_m\cdot e_n\circ T},$$

with the latter expression above estimated using fast Fourier transform of $e_n\circ T$ on a grid eight times finer than the N-grid. Matrix multiplication by $\hat{L}_N$ updates the Fourier coefficients of a function, corresponding to applying L to the function itself; in particular, if $f=\sum_m a_m e_m$, then $\widehat{Lf}_n=\sum_m \hat{L}_{nm}a_m$ provides the updated Fourier coefficients for Lf. See [10] for details.

6.1. Numerical computation of the optimal response of the expectation of an observable

Referring to (10),

(1) We estimate f_0 as the inverse transform of the leading eigenvector $\hat{f}_{N,0}$ of $\hat{L}_N$.
(2) Using the spatial N-grid $0,1/N,2/N,\ldots,(N-1)/N$, we create e_nf_0/T' at these points by pointwise multiplication.
(3) A circular central difference is taken to estimate $(e_nf_0/T')'$.
(4) $L((e_nf_0/T')')$ is estimated in frequency space as $\hat{L}_N((\widehat{e_nf_0/T'})')$.
(5) To numerically estimate $(I-L)^{-1}L((e_nf_0/T')')$ we solve the linear system $\hat{y}_N=(I-\hat{L}_N)\cdot(\hat{L}_N((\widehat{e_nf_0/T'})'))$ and $\hat{y}_{N,1}=0$. The latter condition ensures that the first (constant) Fourier mode is zero, corresponding to mean-zero functions, which is the appropriate subspace for the resolvent $(I-L)^{-1}$ to act on.
(6) The integral with c is performed by taking a dot product of the discrete Fourier transform of the observation $\hat{c}$ and $\hat{y}_N$.
(7) The result is scaled by the appropriate denominator in (10) to produce a_n up to a scaling factor controlled by ν.

In the examples we show results obtained by constraining the perturbation $\dot{T}$ according to the standard Sobolev norm $\|\cdot\|_{H^4}$ and weighted Sobolev norms $\|f\|^2_{H^4,\gamma}=\sum_{i=0}^4\int(f^{(i)}/\gamma^i)^2$, for $\gamma\gg 1$. Increasing γ penalises high derivatives less and allows for optimal $\dot{T}$ with greater irregularity.

6.2. Numerical computation of optimising an isolated spectral value

The isolated spectrum may be estimated by the outer spectrum of $\hat{L}$; see [28] for formal statements. We compute $\hat{L}_N$ as described above and consider one of its eigenvalues λ_0 satisfying $\lambda_0>1/\inf|T_0'|$. Associated with λ_0 is an eigenvector v_0, which we estimate as the inverse Fourier transform of the eigenvector $\hat{v}_0$ of $\hat{L}_N$ corresponding to the eigenvalue λ_0. Referring to (20) we see that we must estimate $L((e_nv_0/T')')$ and the representative ϕ_0 of φ_0 in H^1. The former object is calculated according to steps 1–4 in section 6.1, followed by an inverse FFT. The calculation of the representative of φ_0 is described in the next subsection.

6.2.1. Computing a representative ϕ_0 of φ_0 in H^1.

By the adjoint eigenproperty of φ_0 we have

$$\varphi_0(Lf)=\lambda_0\varphi_0(f)\qquad \text{for all } f\in W^{1,1}. \tag{23}$$

We seek a representative of φ_0 in H^1 and use the ansatz $\phi_0 \approx \sum_{-N/2+1}^{N/2} a_m e_m$. We ask that (23) holds for $f = e_n$, $n = -N/2+1,\dots,N/2$. Thus, using (19) we wish to find $a_m, m = -N/2+1,\dots,N/2$ such that

$$\begin{aligned}
&\int Le_n \cdot \overline{\sum_{m=-N/2+1}^{N/2} a_m e_m} + \int (Le_n)' \cdot \overline{\left(\sum_{m=-N/2+1}^{N/2} a_m e_m\right)'} \\
&\quad = \lambda_0 \left(\int e_n \cdot \overline{\sum_{m=-N/2+1}^{N/2} a_m e_m} + \int (e_n)' \cdot \overline{\left(\sum_{m=-N/2+1}^{N/2} a_m e_m\right)'} \right), \qquad (24)
\end{aligned}$$

for $n = -N/2+1,\dots,N/2$.

Let $L(e_n) \approx \sum_{p=-N/2+1}^{N/2} \hat{L}_{N,pn} e_p$. With this approximation, writing out (24) more explicitly, we ask that

$$\begin{aligned}
0 &= \int (\lambda_0 - L)\, e_n \cdot \overline{\sum_{m=-N/2+1}^{N/2} a_m e_m} + \int [(\lambda_0 - L)\, e_n]' \cdot \overline{\left[\sum_{m=-N/2+1}^{N/2} a_m e_m\right]'} \\
&\approx \int \left(\lambda_0 e_n - \sum_{p=-N/2+1}^{N/2} \hat{L}_{N,pn} e_p\right) \cdot \overline{\sum_{m=-N/2+1}^{N/2} a_m e_m} \\
&\quad + \int \left(\lambda_0 e_n' - \sum_{p=-N/2+1}^{N/2} \left(\hat{L}_{N,pn} e_p\right)'\right) \cdot \overline{\left[\sum_{m=-N/2+1}^{N/2} a_m e_m\right]'} \\
&= \int \left(\lambda_0 e_n - \sum_{p=-N/2+1}^{N/2} \hat{L}_{N,pn} e_p\right) \cdot \overline{\sum_{m=-N/2+1}^{N/2} a_m e_m} \\
&\quad + \int \left(\lambda_0 (2\pi i n)\, e_n - \sum_{p=-N/2+1}^{N/2} \hat{L}_{N,pn} (2\pi i p)\, e_p\right) \cdot \overline{\sum_{m=-N/2+1}^{N/2} a_m (2\pi i m)\, e_m} \\
&= \lambda_0 \bar{a}_n - \sum_{p=-N/2+1}^{N/2} \left(\hat{L}_{N,pn} \bar{a}_p\right) + \lambda_0 (2\pi i n)(-2\pi i n)\, \bar{a}_n \\
&\quad - \sum_{n=-N/2+1}^{N/2} \left(\hat{L}_{N,pn} (2\pi i p)(-2\pi i p)\, \bar{a}_p\right) \\
&= \lambda_0 \left(1 + 4\pi^2 n^2\right) \bar{a}_n - \sum_{p=-N/2+1}^{N/2} \hat{L}_{N,pn} \left(1 + 4\pi^2 p^2\right) \bar{a}_p.
\end{aligned}$$

That is, a satisfies

$$\lambda_0 \bar{a}_n = \sum_{p=-N/2+1}^{N/2} \frac{\hat{L}_{N,pn} \left(1 + (2\pi p)^2\right)}{1 + (2\pi n)^2} \bar{a}_p.$$

In other words, the conjugate a coefficients form a left eigenvector of $\hat{L}_N$, suitably scaled. Since λ_0 and the $\hat{L}_N$ are known numerically, we may easily solve for the a_n, $n = -N/2+1,\ldots,N/2$.

7. Examples

We illustrate theorems 8 and 11 in the following two subsections. In both cases we set $N = 512$.

7.1. Optimising the expectation of observables

We define the expanding circle map $T(x) = 2x - (0.9/2\pi)\sin(2\pi x)$ computed modulo 1. The lower expansivity of T at the fixed point $x = 0$ leads to greater values of the invariant density nearby $x = 0$. A graph of T and its invariant density are shown in figure 1.

Figure 2 illustrates the optimal perturbations $\dot{T}$ for $c(x) = \cos(2\pi x)$ and various weights γ used in the H^4 norm. The map perturbations seek to increase the expectation of c. Because the maximal value of c occurs at $x = 0$ it is advantageous for the perturbations to retain the fixed point at $x = 0$ while simultaneously reducing the expansivity of the map at the fixed point. Such perturbations make the fixed point more 'sticky' and will lead to invariant densities with even greater values at $x = 0$, leading to increases in expectation of c.

Figure 3 carries out the same experiments, replacing the observation function with $c(x) = \sin(2\pi x)$. Now, there is an imperative to remove the sticky fixed point to move invariant mass away from $x = 0$ and toward $x = 0.25$, where c takes its maximum. As shown in figure 3, the strategy is to displace the fixed point by moving it to the right from $x = 0$.

7.2. Optimising the spectrum

To optimise the spectrum, we require an isolated eigenvalue λ_0 of the transfer operator L satisfying $|\lambda_0| > 1/\inf|T_0'|$. We construct a piecewise-linear Markov map of S^1 by linearly connecting the points (rounded to 4 decimal places)

$$x \in \{0.0, 0.1197, 0.2045, 0.2453, 0.3369, 0.3874, 0.49, 0.5875, 0.6336, \\ 0.7343, 0.7695, 0.8523, 1.0\}$$

to their respective images (to be taken mod 1)

$$T_0(x) \in \{0.0, 0.1976, 0.3032, 0.4505, 0.5885, 0.7312, 1.0, 1.1976, \\ 1.3032, 1.4505, 1.5885, 1.7312, 2.0\}$$

as shown in figure 4(left). This Markov map has an isolated spectral value of $\lambda_0 \approx 0.8231$, while $1/\inf|T_0'| \approx 0.6579$; see figure 4(right). We are not aware of another two-branch[8] circle map in the literature whose Perron–Frobenius operator has a positive isolated eigenvalue strictly inside the unit circle, but larger than the reciprocal of the magnitude of minimal slope.

[8] Keller and Rugh [22] construct a two-branch circle map with an isolated negative eigenvalue; by considering two iterates of this map, one would obtain a four-branch circle map with a positive isolated eigenvalue.

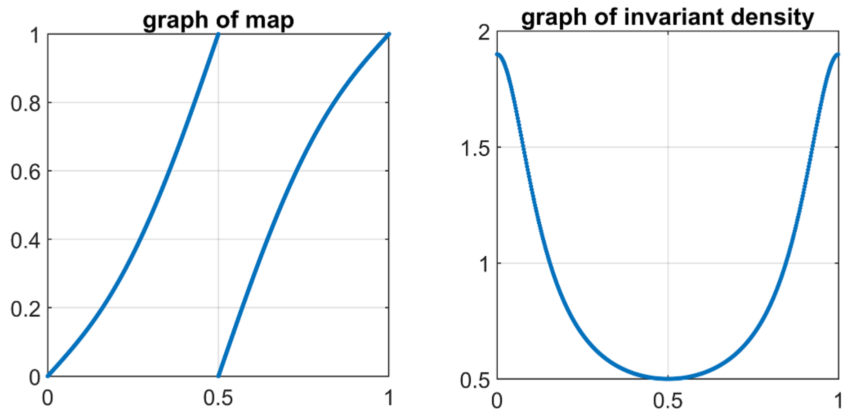

Figure 1. Graph of $x \mapsto 2x - (0.9/2\pi)\sin(2\pi x)$ and corresponding invariant density.

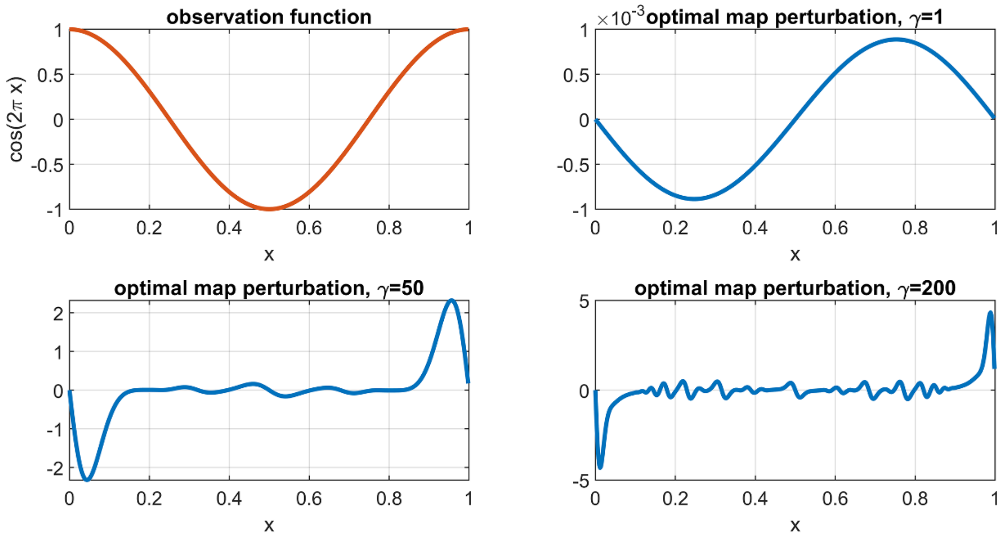

Figure 2. Upper left: observation function $c(x) = \cos(2\pi x)$. Other panels: optimal map perturbations for different weights γ. Each perturbation has unit norm in its corresponding γ-weighted norm. As γ increases, the optimal map perturbation may become more irregular as a smaller penalty is paid for the first to fourth-order derivatives.

We then smooth this map by convolving with a bump function

$$b(x) = \exp\left(\frac{-1}{1-(x/\epsilon)^2}\right)\Big/\kappa\epsilon,$$

using $\epsilon = 1/40$; κ is a constant chosen so that $\int_{S_1} b = 1$. The smoothed map satisfies $1/\inf|T_0'| \approx 0.6579$ and the second largest magnitude eigenvalue of $\hat{L}_N$ for this smoothed map T_0 is $\lambda_0 \approx 0.6992$; as we no longer discuss the original piecewise linear map we reuse the notation T_0 and λ_0. The graph of the smoothed map T_0, its numerical spectrum, and estimates of the eigenvector v_0 and the representative φ_0 are displayed in figure 5.

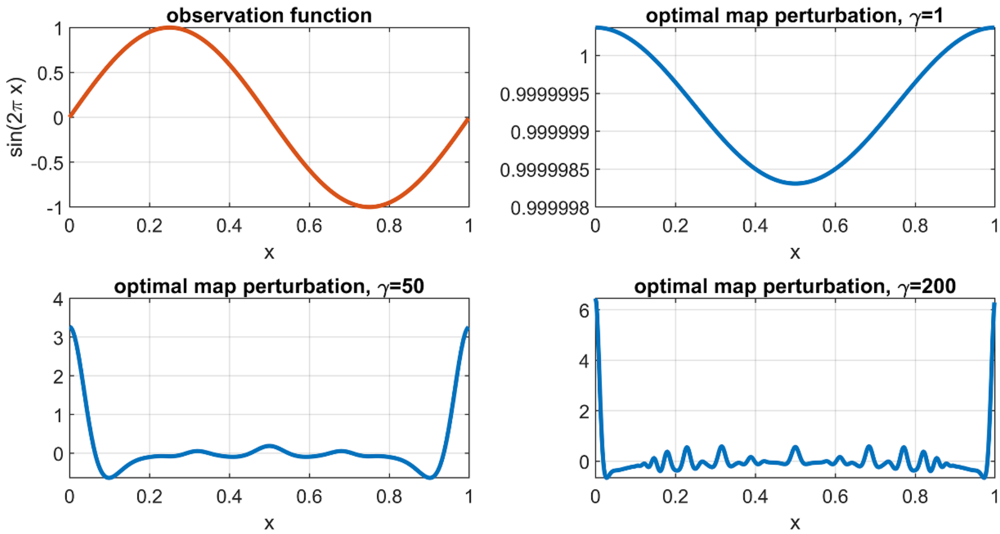

Figure 3. Upper left: observation function $c(x) = \sin(2\pi x)$. Other panels: optimal map perturbations for different weights γ. Each perturbation has unit norm in its corresponding γ-weighted norm. As γ increases, the optimal map perturbation may become more irregular as a smaller penalty is paid for the first to fourth-order derivatives.

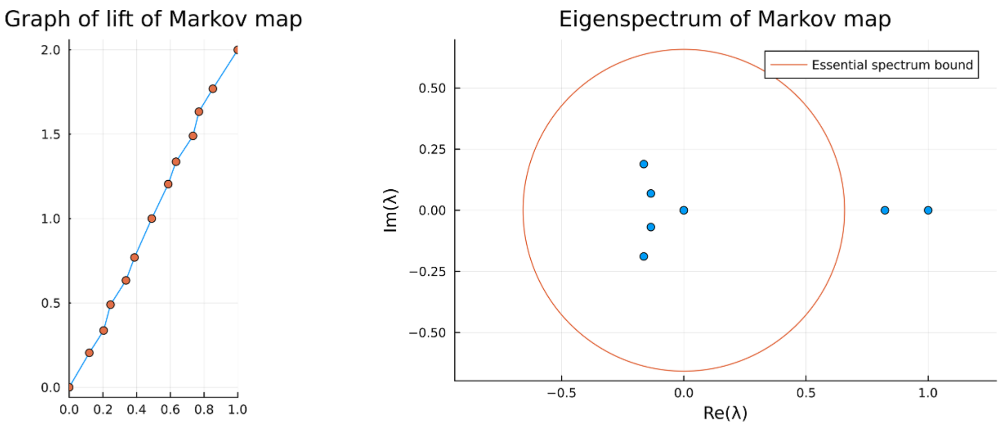

Figure 4. Left: the graph of the lift of the two-full-branch piecewise-linear circle map constructed by joining the 13 point pairs $(x, T_0(x))$ listed toward the beginning of section 7.2 and shown as orange dots in the figure. Right: 12 eigenvalues (6 of which are zero), shown as blue dots, of the 12×12 matrix representing the restriction of the transfer operator of the Markov map to the invariant subspace spanned by 12 indicator functions supported on the domains of linearity of the map. The essential spectrum bound produced by the inverse of the minimum slope of the 12 linear branches is shown in red.

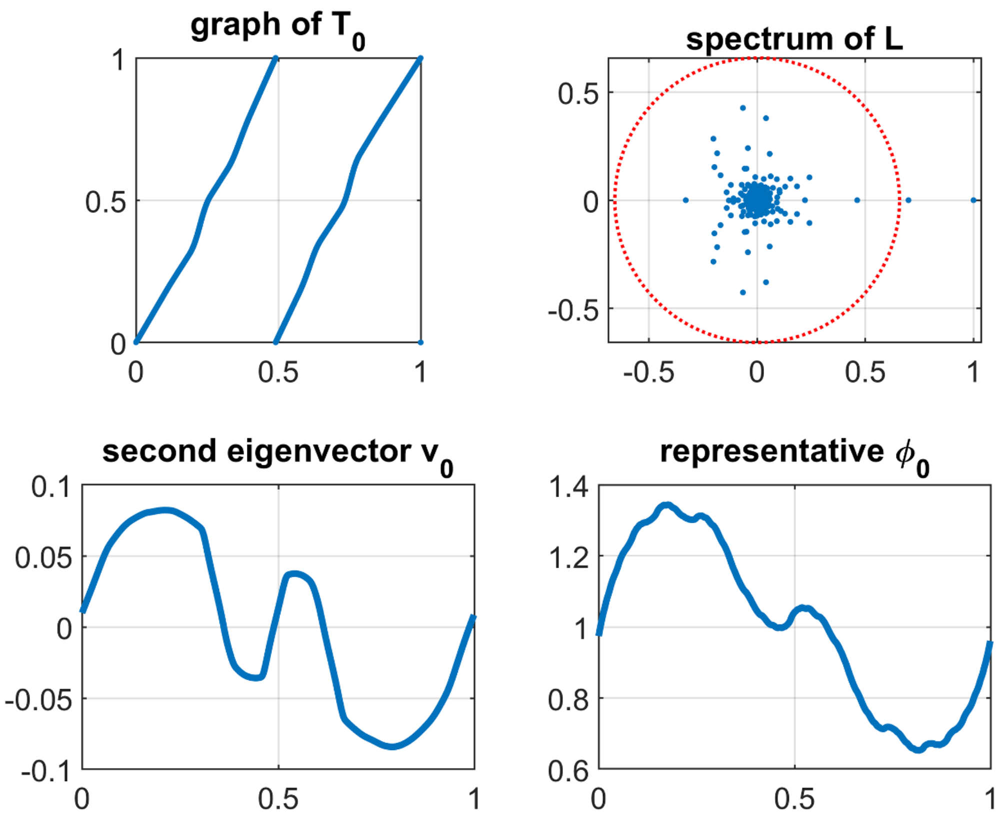

Figure 5. Upper left: graph of a smoothed T_0 derived from the map in figure 4(left). Upper right: spectrum of L_N, with $1/\inf|T_0'|$ indicated as a dotted red circle. Lower left: second eigenvector u_0, normalised so that $\varphi_0(u_0) = 1$. Lower right: representative of φ_0 in $W^{1,\infty}$, normalised so that $\varphi_0(\mathbf{1}) = 1$.

Figure 6 illustrates the optimal perturbations $\dot{T}$ to maximally increase the isolated spectral value. Using theorem 11 we may also compute the linear responses of λ_0 with respect to the optimal map perturbations $\dot{T}$ shown in figure 6 for various γ-weighted H^4 norms. For $\gamma = 1, 25, 50$, and 200 we find that $\dot{\lambda}(\dot{T})$ is approximately 0.5758, 3.1427, 11.1590, and 125.51, respectively. These values indicate that as we reduce the penalty on the irregularity of the perturbations $\dot{T}$ by increasing γ, we may increase the corresponding linear response of λ_0 without limit. To put these numbers in perspective, we remark that even a movement of the spectral value λ_0 by an amount 0.1 would be dramatic, and with $\gamma = 200$, making a macroscopic peturbation of T_0 by $\dot{T}/1000$ would exceed such a movement (up to linear approximation).

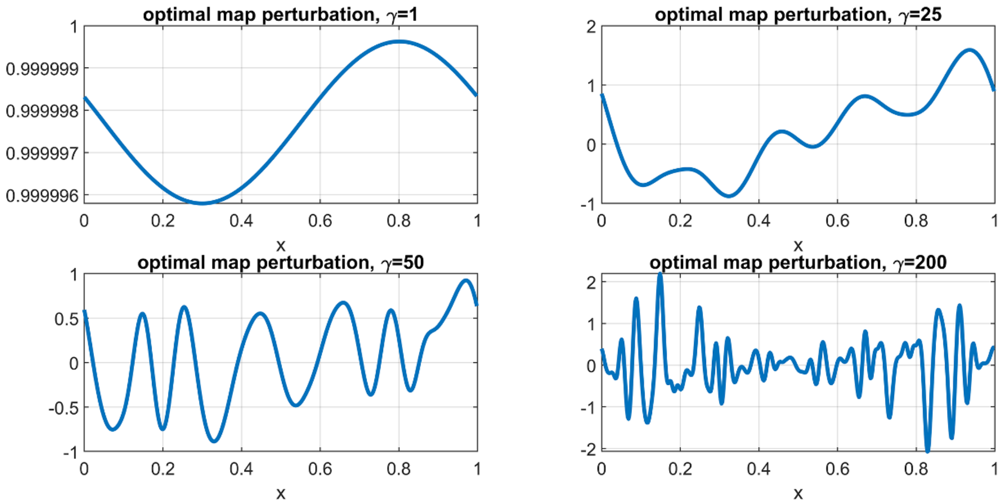

Figure 6. Optimal perturbations $\dot{T}$ for different weights γ. Each perturbation $\dot{T}$ has unit norm in its corresponding γ-weighted norm. As γ increases, the optimal map perturbation may become more irregular because a smaller penalty is paid for the first to fourth-order derivatives. The values of $\dot{\lambda}(\dot{T})$ are (in order upper left, upper right, lower left, lower right): 0.5758, 3.1427, 11.1590, and 125.51, demonstrating that as we allow greater irregularity in the γ-weighted norm we are able to increase the spectral response.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Acknowledgments

GF thanks the University of Pisa's Department of Mathematics for its generous hospitality during a visit that initiated this research. GF is partially supported by an Australian Research Council Discovery Project. SG is partially supported by the research project 'Stochastic properties of dynamical systems' (PRIN 2022NTKXCX) of the Italian Ministry of Education and Research.

Conflicts of interests

This submission raises no conflicts of interests or competing interests for the authors.

Appendix. General linear response formulas for eigenvalues and eigenvectors and application to expanding maps

In this section we recall general results for the linear response of fixed points, eigenvalues and eigenvectors of Markov operators under suitable perturbations. We then develop the estimates that are necessary to apply the results to expanding maps and deterministic perturbations. Let X be a compact Riemann manifold and let m be its normalised volume measure. Denote by $L^1(X,m)$ or simply by L^1 the space of m-integrable functions. We consider a sequence of Banach spaces

$$(B_{ss}, \| \|_{ss}) \subseteq (B_s, \| \|_s) \subseteq (B_w, \| \|_w),$$

with $B_w \subseteq L^1(X,m)$ and the norms satisfying

$$\| \|_w \leqslant \| \|_s \leqslant \| \|_{ss}.$$

Let us suppose that B_{ss} contains the constant functions. For $i \in \{ss, s, w\}$ define the following (closed) zero-mean function spaces $V_{ss} \subseteq V_s \subseteq V_w$ by $V_i := \{f \in B_i : \int_X f \, \mathrm{d}m = 0\}$. We will consider Markov[9] operators acting on these spaces. If A, B are two normed vector spaces and $T : A \to B$ we denote the mixed norm $\|T\|_{A \to B}$ as

$$\|T\|_{A\to B} := \sup_{f \in A, \|f\|_A \leqslant 1} \|Tf\|_B.$$

A.1. Abstract linear response of invariant densities

Under the general assumptions and notations above we are now going to state an abstract result on the linear response of invariant densities. Similar results and constructions appear in the literature, applied to specific classes of examples, we include a proof of the statement for completeness.

Theorem 12. *Let us consider* $\overline{\delta} > 0$, $\delta \in [0, \overline{\delta})$ *and a family of Markov operators* $L_\delta : B_w \to B_w$. *Suppose that for* $\delta \in [0, \overline{\delta})$ *there is a probability density* $v_\delta \in B_s$ *such that*

$$L_\delta v_\delta = v_\delta$$

and that there is $\dot{L}v_0 \in B_s$ *such that*

$$\lim_{\delta \to 0} \left\| \frac{L_\delta - L_0}{\delta} v_0 - \dot{L}v_0 \right\|_s = 0. \tag{25}$$

Suppose that for $\delta \in [0, \overline{\delta})$ *the resolvent operators* $(Id - L_\delta)^{-1}$ *of* L_δ *are defined and bounded from* V_s *to* V_w,

$$|| (Id - L_\delta)^{-1} ||_{V_s \to V_w} < +\infty \tag{26}$$

and

$$\lim_{\delta \to 0} || (Id - L_\delta)^{-1} - (Id - L_0)^{-1} ||_{V_s \to V_w} = 0. \tag{27}$$

Then

$$\lim_{\delta \to 0} \left\| \frac{v_\delta - v_0}{\delta} - (Id - L_0)^{-1} \dot{L}v_0 \right\|_{V_w} = 0.$$

[9] A Markov operator $L : B \to B$ is a linear operator satisfying (i) $L(f) \geqslant 0$ if $f \geqslant 0$ and (ii) $\int_X L(f)\, dm = \int_X f\, dm$ for all $f \in B$.

Proof. We have that for each $\delta \in [0,\overline{\delta})$, v_δ is a fixed point of L_δ. Using this we get

$$\begin{aligned}(Id - L_\delta)\frac{v_\delta - v_0}{\delta} &= \frac{v_\delta - v_0}{\delta} - \frac{L_\delta v_\delta - L_\delta v_0}{\delta}\\ &= \frac{-v_0 + L_\delta v_0}{\delta}\\ &= \frac{1}{\delta}(L_\delta - L_0)v_0.\end{aligned}$$

We remark that for each δ, L_δ preserves V_s. Since $\forall\delta > 0$, $\frac{L_\delta - L_0}{\delta}v_0 \in V_s$ and by the assumptions, (26) and (27), for δ small enough $(Id - L_\delta)^{-1} : V_s \to V_w$ is a uniformly bounded operator. We can apply the resolvent to both sides of the expression above to get

$$\begin{aligned}(Id - L_\delta)^{-1}(Id - L_\delta)\frac{v_\delta - v_0}{\delta} &= (Id - L_\delta)^{-1}\frac{L_\delta - L_0}{\delta}v_0\\ &= (Id - L_\delta)^{-1}\frac{L_\delta - L_0}{\delta}v_0 - (Id - L_0)^{-1}\frac{L_\delta - L_0}{\delta}v_0\\ &\quad + (Id - L_0)^{-1}\frac{L_\delta - L_0}{\delta}v_0.\end{aligned} \tag{28}$$

Since $||(Id - L_\delta)^{-1} - (Id - L_0)^{-1}||_{V_s \to V_w} \to 0$ we have

$$\left\|\left[(Id - L_\delta)^{-1} - (Id - L_0)^{-1}\right]\frac{L_\delta - L_0}{\delta}v_0\right\|_w \leqslant \left\|(Id - L_\delta)^{-1} - (Id - L_0)^{-1}\right\|_{V_s \to V_w}\left\|\frac{L_\delta - L_0}{\delta}v_0\right\|_s \to 0.$$

Since $\lim_{\delta\to 0}\frac{L_\delta - L_0}{\delta}v_0$ converges in V_s, then (28) implies that in the B_w topology

$$\begin{aligned}\lim_{\delta\to 0}\frac{v_\delta - v_0}{\delta} &= \lim_{\delta\to 0}(Id - L_0)^{-1}\frac{L_\delta - L_0}{\delta}v_0\\ &= (Id - L_0)^{-1}\lim_{\delta\to 0}\frac{L_\delta - L_0}{\delta}v_0\\ &= (Id - L_0)^{-1}\left[\dot{L}v_0\right].\end{aligned}$$

□

A.2. Abstract linear response of the resolvent and linear response of eigenvalues

In this section we provide general statements about the response of eigenvalues and resolvent operators. The results presented follow from classical statements we adapt to our purposes. Corollary 14 illustrates the abstract linear response of the resolvent and proposition 15 provides differentiability of an isolated eigenvalue and of its corresponding eigenvector in B_w.

We recall the abstract linear response result for the resolvent operator proved in [19] and we apply it to get linear response formulas for simple eigenvalues and eigenvectors of transfer operators. Recalling the Banach spaces B_{ss}, B_s, B_w from the last subsection, let us suppose that all are separable and that B_{ss} is dense in B_s. Let us consider $\overline{\delta} > 0$, $\delta \in [0,\overline{\delta})$ and a family of Markov operators L_δ and an operator $\dot{L} : B_s \to B_w$ satisfying the following assumptions: there are $C \geqslant 0$ and $0 < \alpha < 1$ such that for each $n \in \mathbb{N}$ and $\delta \in [0,\overline{\delta})$:

(GL1) $||L_\delta^n||_w \leqslant C||f||_w$
(GL2) $||L_\delta^n||_s \leqslant C\alpha^n||f||_s + C||f||_w$, $||L_\delta^n||_{ss} \leqslant C\alpha^n||f||_{ss} + C||f||_s$
(GL3) $||L_\delta||_{B_s \to B_s} \leqslant C$, and $||L_\delta||_{B_{ss} \to B_{ss}} \leqslant C$

(GL4) $||L_\delta - L_0||_{B_s \to B_w} \leqslant C\delta$ and $||L_\delta - L_0||_{B_{ss} \to B_s} \leqslant C\delta$.
(GL5) $||\dot{L}||_{B_s \to B_w} \leqslant C$, $||\dot{L}||_{B_{ss} \to B_s} \leqslant C$ and
(GL6) $||L_\delta - L_0 - \delta\dot{L}||_{B_{ss} \to B_w} \leqslant C\delta^2$.

Under these assumptions one has:

Theorem 13 ([19], Thm. 8.1). *Consider a family of operators L_δ satisfying the assumptions (GL1),...,(GL6). Further consider $\delta' > 0$, $\alpha' > \alpha > 0$ and the set*

$$Q_{\delta',\alpha'} = \{z \in \mathbb{C} : |z| \geqslant \alpha', dist(\sigma(L_0), z) > \delta'\}$$

where $\sigma(L_0) = \sigma_s(L_0) \cup \sigma_{ss}(L_0)$ and $\sigma_s(L_0), \sigma_{ss}(L_0)$ denote the spectrum of L_0 acting on B_s and B_{ss} respectively.

Then there is $C_2 \geqslant 0$ such that for each $z \in Q_{\delta',\alpha'}$ and δ small enough

$$|| (z - L_\delta)^{-1} - (z - L_0)^{-1} \left(Id + \delta\dot{L}(z - L_0)^{-1}\right) ||_{B_{ss} \to B_w} \leqslant C_2|\delta|^{1+\eta} \tag{29}$$

where $\eta = \log(\alpha'/\alpha)/\log(1/\alpha)$.

We remark that (29) gives the first-order change of the resolvent when the operator is perturbed; in fact from (29) one has the following immediate rearrangement

$$||(z - L_\delta)^{-1} - (z - L_0)^{-1} + \delta(z - L_0)^{-1}\dot{L}(z - L_0)^{-1})||_{B_{ss} \to B_w} \leqslant C_2|\delta|^{1+\eta} \tag{30}$$

and the following corollary:

Corollary 14. *Under the above assumptions and with the same notations*

$$\left\| \frac{(z - L_\delta)^{-1} - (z - L_0)^{-1}}{\delta} - (z - L_0)^{-1}\dot{L}(z - L_0)^{-1} \right\|_{B_{ss} \to B_w} \leqslant C_2|\delta|^{\eta}.$$

We will also make the following assumption on the spaces involved, which is well known to imply together with the assumptions (GL1) and (GL2), the quasicompactness of the transfer operators L_δ when acting on B_s and on B_{ss}.

(GL7) B_s is compactly immersed in B_w and B_{ss} is compactly immersed in B_s.

Let $\mathbf{1}$ be the density representing the indicator function of S^1. Let λ_0 be a simple isolated eigenvalue for L_0 acting on B_s. Lemma III.3 of [20] now ensures the existence of a unique eigenfuction $\varphi_0 \in B_s^*$ of the adjoint operator $L_0^* : B_s^* \to B_s^*$ corresponding to λ_0 ($L_0^*\varphi_0 = \lambda_0\varphi_0$), scaled so that $\varphi_0(\mathbf{1}) = 1$. Suppose λ_δ is an isolated, simple eigenvalue for L_δ acting on B_s. Suppose v_δ is an eigenvector of L_δ associated to λ_δ. To quantitatively address the differentiability of v_δ and λ_δ we need to scale v_δ consistently. We will rescale v_δ in a way that $\varphi_0(v_\delta) = 1$ for all $\delta \in [0, \overline{\delta}]$. Let us consider a simple eigenvalue $\lambda_\delta \in \sigma(L_\delta)$, and $\theta > 0$ such that

$$\{|z - \lambda_\delta| \leqslant \theta\} \cap \sigma(L_\delta) = \{\lambda_\delta\}.$$

The eigenprojection operator $\Pi_{\lambda_\delta} : B_s \to B_s$ is defined by

$$\Pi_{\lambda_\delta} := \frac{1}{2\pi i}\int_{\{|z-\lambda_\delta|=\theta\}} (z \cdot Id - L_\delta)^{-1}\, \mathrm{d}z. \tag{31}$$

It is well known that this is a projection ($\Pi_{\lambda_\delta}^2 = \Pi_{\lambda_\delta}$) and it does not depend on θ or on the circle $\{|z-\lambda_\delta| = \theta\}$, which can be replaced by any smooth simple curve only containing the simple eigenvalue λ_δ. Furthermore $L_\delta = \Pi_{\lambda_\delta} L_\delta \Pi_{\lambda_\delta} + (Id - \Pi_{\lambda_\delta}) L_\delta (Id - \Pi_{\lambda_\delta})$ and $\sigma(\Pi_{\lambda_\delta} L_\delta \Pi_{\lambda_\delta}) = \{0, \lambda_\delta\}$. Hence Π_{λ_δ} is a projection on the λ_δ-eigenspace of L_δ.

We now consider the dependence of the isolated eigenvalue λ_δ on δ. First we consider the associated eigenvector v_δ and state a formula for its derivative with respect to δ.

Proposition 15. *Suppose the family of transfer operators L_δ satisfy the assumptions (GL1),...,(GL7). Suppose $|\lambda_0| > \alpha$ (see (GL2)) is a simple isolated eigenvalue for L_0 acting on B_s. Then*

(I) λ_0 is also an eigenvalue of the operator applied to B_{ss}. Furthermore, for δ small enough L_δ has a family of simple isolated eigenvalues (both for the operator applied to B_s and B_{ss}) λ_δ with $\lambda_\delta \to \lambda_0$ as $\delta \to 0$.

(II) each λ_δ has an eigenvector $v_\delta \in B_s$, rescaled by $\varphi_0(v_\delta) = 1$ as described before, and as $\delta \to 0$

$$\lim_{\delta\to 0} ||v_0 - v_\delta||_s = 0. \tag{32}$$

Further, one has

$$\lim\nolimits_{\delta\to 0} \frac{v_\delta - v_0}{\delta} = \dot{v} := \frac{1}{2\pi i} \int_{\{|z-\lambda_0|=\theta\}} (z - L_0)^{-1} \dot{L} (z - L_0)^{-1} v_0 \mathrm{d}z$$

with convergence in B_w. Moreover, the function $\delta \to \lambda_\delta$ is differentiable at $\delta = 0$.

Proof. Lemma A.3 of [4] (following a similar result proved in [5]) implies that if (i) B_s is separable and $L : B_s \to B_s$ is a continuous linear map preserving a dense continuously embedded subspace B_{ss}, (ii) $L : B_{ss} \to B_{ss}$ is continuous and (iii) the essential spectral radius of L considered both as acting on B_s and on B_{ss} is bounded by $0 < \rho < 1$, then the simple eigenvalues of $L : B_s \to B_s$ and $L : B_{ss} \to B_{ss}$ in $\{z \in \mathbb{C} |\ ||z|| > \rho\}$ coincide, the associated eigenspaces also coincide and are contained in B_{ss}. The assumptions (GL1),...,(GL3) and (GL7) imply that the essential spectral radius is bounded by α (see e.g. lemma 2.2 of [6]). The assumptions (GL1),...,(GL3) also imply the required continuity properties for the operators acting on B_s and B_{ss}, thus we can apply lemma A.3 of [4] and establish that λ_0 is a simple eigenvalue of L_0 applied on B_{ss} and the associated eigenspace is generated by an eigenvector $v_0 \in B_{ss}$. As a classical consequence of the assumptions (GL1),...,(GL4) and (GL7) the Keller–Liverani spectral stability theorem (see [21], theorem 1). establishes that for δ small enough, λ_δ is simple and the associated eigenspaces are generated by $v_\delta \in B_s$. Furthermore one has that $\lim_{\delta\to 0} |\lambda_\delta - \lambda_0| = 0$ and $||\Pi_{\lambda_0} - \Pi_{\lambda_\delta}||_{B_{ss}\to B_s} \to 0$. Since $\Pi_{\lambda_0}(v_0) = v_0$, $\varphi_0(\Pi_{\lambda_0}(v_0)) = 1$ and $||\Pi_{\lambda_0}(v_0) - \Pi_{\lambda_\delta}(v_0)||_s \to 0$, then $v_\delta = \frac{\Pi_{\lambda_\delta}(v_0)}{\varphi_0(\Pi_{\lambda_\delta}(v_0))}$ varies continuously in B_s at $\delta = 0$. This takes care of part (I) and part (II) up to equation (32).

For the remainder of part (II), putting together (30) and (31) we have

$$\begin{aligned}
\Pi_{\lambda_\delta} &= \frac{1}{2\pi i} \int_{\{|z-\lambda_\delta|=\theta\}} (z - L_\delta)^{-1} \mathrm{d}z \\
&= \frac{1}{2\pi i} \int_{\{|z-\lambda_\delta|=\theta\}} (z - L_0)^{-1} + \delta (z - L_0)^{-1} \dot{L} (z - L_0)^{-1} + O_\delta \,\mathrm{d}z
\end{aligned}$$

where O_δ represents an operator such that there is a constant $\tilde{C}$ so that $\frac{||O_\delta||_{B_{ss}\to B_w}}{\delta} \leqslant \tilde{C}|\delta|^\eta$ for sufficiently small δ (here η is the one defined in theorem 13). Thus, since for δ small enough, $\{|z-\lambda_\delta|=\theta\}$ also encircles λ_0

$$\begin{aligned}\Pi_{\lambda_\delta} &= \Pi_{\lambda_0} + \delta\frac{1}{2\pi i}\int_{\{|z-\lambda_\delta|=\theta\}}(z-L_0)^{-1}\dot{L}(z-L_0)^{-1}\,\mathrm{d}z + \frac{1}{2\pi i}\int_{\{|z-\lambda|=\theta\}} O_\delta\ \mathrm{d}z \\ &= \Pi_{\lambda_0} + \delta\frac{1}{2\pi i}\int_{\{|z-\lambda_\delta|=\theta\}}(z-L_0)^{-1}\dot{L}(z-L_0)^{-1}\,\mathrm{d}z + O_{2,\delta}\end{aligned}$$

where $\frac{||O_{2,\delta}||_{B_{ss}\to B_w}}{\delta} \to 0$ as $\delta \to 0$.

Since, as proved above, $v_0 \in B_{ss}$, we then have

$$\begin{aligned} v_\delta &= \frac{\Pi_{\lambda_\delta}(v_0)}{\varphi_0(\Pi_{\lambda_\delta}(v_0))} \\ &= \frac{\Pi_{\lambda_0}(v_0) + \delta[\frac{1}{2\pi i}\int_{\{|z-\lambda_\delta|=\theta\}}(z-L_0)^{-1}\dot{L}(z-L_0)^{-1}v_0\mathrm{d}z] + q_\delta}{\varphi_0(\Pi_{\lambda_\delta}(v_0))} \\ &= \frac{v_0 + \delta[\frac{1}{2\pi i}\int_{\{|z-\lambda_\delta|=\theta\}}(z-L_0)^{-1}\dot{L}(z-L_0)^{-1}v_0\mathrm{d}z] + q_\delta}{\varphi_0(\Pi_{\lambda_\delta}(v_0))}\end{aligned}$$

where $\frac{||q_\delta||_w}{\delta} \to 0$ and using the fact $||\Pi_{\lambda_0} - \Pi_{\lambda_\delta}||_{B_{ss}\to B_s} \to 0$, one has $\varphi_0(\Pi_{\lambda_\delta}(v_0)) \to 1$. Since $\lim_{\delta\to 0}|\lambda_\delta - \lambda_0| = 0$ we have for every fixed θ small enough

$$\lim_{\delta\to 0}\int_{\{|z-\lambda_\delta|=\theta\}}(z-L_0)^{-1}\dot{L}(z-L_0)^{-1}v_0\mathrm{d}z = \int_{\{|z-\lambda_0|=\theta\}}(z-L_0)^{-1}\dot{L}(z-L_0)^{-1}v_0\mathrm{d}z.$$

We then get:

$$\frac{v_\delta - v_0}{\delta} \to \frac{1}{2\pi i}\int_{\{|z-\lambda_0|=\theta\}}(z-L_0)^{-1}\dot{L}(z-L_0)^{-1}v_0\mathrm{d}z$$

in the B_w topology.

For the differentiability of $\delta \mapsto \lambda_\delta$, let us consider the normalised eigenvector v_δ as used before. Using that $L_\delta(v_\delta) = \lambda_\delta v_\delta$ we get

$$\frac{L_\delta v_\delta - L_0 v_0}{\delta} = \frac{\lambda_\delta v_\delta - \lambda_0 v_0}{\delta}.$$

Thus

$$L_\delta\frac{v_\delta - v_0}{\delta} + \frac{L_\delta - L_0}{\delta}v_0 = \lambda_\delta\frac{v_\delta}{\delta} - \lambda_0\frac{v_0}{\delta} = \lambda_\delta\frac{v_\delta - v_0}{\delta} + v_0\frac{\lambda_\delta - \lambda_0}{\delta}. \tag{33}$$

To conclude we will argue that both terms on the LHS of (33) and the first term on the RHS of (33) converge in the B_w topology; this implies that the second term on the RHS of (33) also converges, yielding differentiability of $\delta \mapsto \lambda_\delta$. Regarding the first term on the LHS of (33), by (GL4) and (32) we have that $\lim_{\delta\to 0}||L_\delta\frac{v_\delta - v_0}{\delta} - L_0\frac{v_\delta - v_0}{\delta}||_w = 0$; thus we may replace L_δ with L_0 in this term. Furthermore, $\frac{L_\delta - L_0}{\delta}v_0$ converges in B_w by (GL6) and $\frac{v_\delta - v_0}{\delta}$ converges in B_w as proved above. Comparing the LHS and RHS of (33) we see that $\lim_{\delta\to 0}\frac{\lambda_\delta - \lambda_0}{\delta}$ exists. □

A.3. An abstract formula for the linear response of the spectrum

Proposition 16 constructs the abstract formula for the derivative of the eigenvalue.

Proposition 16. *Suppose the family of transfer operators L_δ satisfy the assumptions (GL1),...,(GL7). Suppose λ_0 with $|\lambda_0| > \alpha$ is an isolated eigenvalue for L_0 acting on B_s and that there is $\dot{L}v_0 \in B_s$ such that*

$$\lim_{\delta\to 0}\left\|\frac{L_\delta - L_0}{\delta}v_0 - \dot{L}v_0\right\|_s = 0 \tag{34}$$

as in (25).

Then we have the following response formula for the simple isolated eigenvalue

$$\dot{\lambda} := \lim_{\delta\to 0}\frac{\lambda_\delta - \lambda_0}{\delta} = \varphi_0\left(\dot{L}v_0\right).$$

Proof. We prove $\lim_{\delta\to 0}\left|\frac{\lambda_\delta-\lambda_0}{\delta} - \varphi_0(\dot{L}v_0)\right| = 0$. Write

$$\left|\frac{\lambda_\delta - \lambda_0}{\delta} - \varphi_0\left(\dot{L}v_0\right)\right| \leqslant \left|\frac{\lambda_\delta - \lambda_0}{\delta} - \frac{\varphi_0\left(L_\delta - L_0\right)\left(v_0\right)}{\delta}\right| + \left|\frac{\varphi_0\left(L_\delta - L_0\right)\left(v_0\right)}{\delta} - \varphi_0\left(\dot{L}v_0\right)\right|.$$

By the assumption (34) and the fact that $\varphi_0 \in B_s^*$ we have $\left|\frac{\varphi_0(L_\delta - L_0)(v_0)}{\delta} - \varphi_0(\dot{L}v_0)\right| \to 0$.

Let us now consider $\lim_{\delta\to 0}\left|\frac{\lambda_\delta-\lambda_0}{\delta} - \frac{\varphi_0(L_\delta-L_0)(v_0)}{\delta}\right|$. Recalling that by the normalisations introduced before (see the first paragraphs after corollary 14) we have $\varphi_0(v_\delta) = 1$ for $\delta \in [0,\overline{\delta}]$, and $\varphi_0(L_0(v_0)) = \lambda_0\varphi_0(v_0) = \lambda_0$ and $\varphi_0(L_0(v_\delta)) = \lambda_0$. We then can write

$$\begin{aligned}\left|\frac{\lambda_\delta - \lambda_0}{\delta} - \frac{\varphi_0\left(L_\delta - L_0\right)\left(v_0\right)}{\delta}\right| &= \left|\frac{\lambda_\delta - \lambda_0}{\delta} - \frac{\varphi_0\left(L_\delta\left(v_0\right)\right) - \lambda_0}{\delta}\right| = \left|\frac{\lambda_\delta - \varphi_0\left(L_\delta\left(v_0\right)\right)}{\delta}\right| \\ &= \left|\frac{\varphi_0\left(L_\delta\left(v_\delta\right)\right) - \varphi_0\left(L_\delta\left(v_0\right)\right) - \left[\varphi_0\left(L_0\left(v_\delta\right)\right) - \varphi_0\left(L_0\left(v_0\right)\right)\right]}{\delta}\right| \\ &= \left|\frac{\varphi_0\left(\left[L_\delta - L_0\right]\left(v_\delta - v_0\right)\right)}{\delta}\right|.\end{aligned}$$

We remark that since φ_0 is a left eigenvector

$$\varphi_0\left(\left[L_\delta - L_0\right]\left(v_\delta - v_0\right)\right) = \varphi_0\left(\lambda_0^{-n}L_0^n\left(\left[L_\delta - L_0\right]\left(v_\delta - v_0\right)\right)\right) \tag{35}$$

and by (GL2)

$$||L_0^n\left(\left[L_\delta - L_0\right]\left(v_\delta - v_0\right)\right)||_s \leqslant C\alpha^n||\left[L_\delta - L_0\right]\left(v_\delta - v_0\right)||_s + C||\left[L_\delta - L_0\right]\left(v_\delta - v_0\right)||_w. \tag{36}$$

We are now going to estimate $||[L_\delta - L_0](v_\delta - v_0)||_s$ and $||[L_\delta - L_0](v_\delta - v_0)||_w$.

We remark that $\forall\delta \in [0,\overline{\delta})$ and $n \geqslant 0$ by the fact that v_δ is an eigenvector, and by (GL2)

$$||v_\delta||_{ss} = ||\lambda_\delta^{-n}L_\delta^n\left(v_\delta\right)||_{ss} \leqslant C|\lambda_\delta^{-n}|\alpha^n||v_\delta||_{ss} + C|\lambda_\delta^{-n}|\ ||v_\delta||_s.$$

Since we have $\lambda_\delta \to \lambda_0$ and $\alpha < |\lambda_0|$ we can choose $\hat{\delta} < \overline{\delta}$ and n such that $C|\lambda_\delta^{-n}|\alpha^n < \frac{1}{2}$ and $|\lambda_\delta^{-n}| < \alpha^{-n}$ for each $\delta \in [0,\hat{\delta})$.

We then have that for this choice of n, $\forall\delta \in [0,\hat{\delta})$

$$||v_\delta||_{ss} \leqslant 2C\alpha^{-n}||v_\delta||_s$$

and since by (32) $||v_\delta||_s$ is uniformly bounded, then also $||v_\delta||_{ss}$ is uniformly bounded for $\delta \in [0, \hat{\delta})$. By this $||v_\delta - v_0||_{ss}$ is also uniformly bounded, thus by (GL4)

$$||\left[L_\delta - L_0\right](v_\delta - v_0)||_s \leqslant \delta C \sup_{\delta \in [0,\hat{\delta})} ||v_\delta - v_0||_{ss}. \tag{37}$$

Again, by (GL4)

$$||\left[L_\delta - L_0\right](v_\delta - v_0)||_w \leqslant \delta C ||v_\delta - v_0||_s$$

and since $||v_\delta - v_0||_s \to 0$ we have that

$$\delta^{-1} ||\left[L_\delta - L_0\right](v_\delta - v_0)||_w \to 0. \tag{38}$$

Let us fix some $\epsilon > 0$, by (37), there is n such that $\forall\ \delta \in [0, \hat{\delta})$

$$C\lambda_0^{-n}\alpha^n ||\left[L_\delta - L_0\right](v_\delta - v_0)||_s \leqslant \delta \frac{\epsilon}{2}.$$

Once n is fixed, by (38) there is $\hat{\delta}' \leqslant \hat{\delta}$ such that $\forall\ \delta \in [0, \hat{\delta}')$

$$C\lambda_0^{-n} ||\left[L_\delta - L_0\right](v_\delta - v_0)||_w \leqslant \delta \frac{\epsilon}{2}.$$

Now we are ready to prove that $\frac{|\varphi_0([L_\delta - L_0](v_\delta - v_0))|}{\delta} \overset{\delta\to 0}{\to} 0$. By (35) and (36) we can bound $\frac{|\varphi_0([L_\delta - L_0](v_\delta - v_0))|}{\delta}$ as follows: choosing n as above, for each $\delta \in [0, \hat{\delta}')$ we get

$$\begin{aligned}
\delta^{-1} |\varphi_0 ([L_\delta - L_0](v_\delta - v_0))| &= \delta^{-1} |\varphi_0 \left(\lambda_0^{-n} L_0^n ([L_\delta - L_0](v_\delta - v_0))\right)| \\
&\leqslant \delta^{-1} ||\varphi_0||_{B_s^*} ||\lambda_0^{-n} L_0^n ([L_\delta - L_0](v_\delta - v_0))||_s \\
&\leqslant ||\varphi_0||_{B_s^*} \delta^{-1} \left[\delta \frac{\epsilon}{2} + \delta \frac{\epsilon}{2}\right] \\
&\leqslant \epsilon ||\varphi_0||_{B_s^*}
\end{aligned}$$

which, since ϵ is arbitrary, can be made as small as wanted as $\delta \to 0$, proving the statement. □

A.4. Verifying abstract transfer operator conditions using properties of expanding maps

In this section we develop more explicit estimates that allow us to apply the above abstract theory to expanding maps and suitable perturbations (verifying (A0)–(A2)). This will lead to the proof of proposition 3. In the following we will then consider as a stronger and weaker spaces B_{ss}, B_s, B_w considered above, the spaces of Borel densities in a Sobolev space $W^{3,1}, W^{1,1}, L^1$. The transfer and derivative operators associated to expanding maps will be defined as acting on densities as in definition 1. In the following we will recall some basic facts on the properties of such operators.

A.4.1. Uniform estimates for individual maps. Assumptions (GL1)–(GL3) require the transfer operators to satisfy uniform norm and Lasota–Yorke estimates. These hypotheses will hold when the operators are associated to a uniform family of maps.

Definition 17. A set $U_{M,N}$ of expanding maps of S^1 is called a *uniform* C^k*family* with parameters $M \geqslant 0$ and $N > 1$ if it satisfies uniformly the expansiveness and regularity condition: $\forall T \in U_{M,N}$

$$||T||_{C^k} \leqslant M, \inf_{x \in S^1} |T'(x)| \geqslant N.$$

It is well known that the transfer operator associated to a smooth expanding map has some regularisation properties when acting on suitable Sobolev spaces (see e.g. [25] and [18]). This is expressed in the following Lemma.

Lemma 18 ([25], section 1.5; [18], lemma 29). *Let $k \geqslant 2$ and let $U_{M,N}$ be a uniform C^k family of expanding maps of S^1. The transfer operators L_T associated to each $T \in U_{M,N}$ satisfy a uniform Lasota-Yorke inequality on $W^{i,1}(S^1)$: let $\alpha = N^{-1} < 1$. For each $1 \leqslant i \leqslant k-1$ there are A_i, $B_i \geqslant 0$ such that for each $n \geqslant 0$, $T \in U_{M,N}$*

$$\begin{aligned} &\|L_T^n\|_{W^{i-1,1}} \leqslant A_i \\ &||L_T^n f||_{W^{i,1}} \leqslant \alpha^{ni}\|f\|_{W^{i,1}} + B_i\|f\|_{W^{i-1,1}} \qquad \text{for all } f \in W^{i,1}. \end{aligned}$$

Lemma 18 allow us to establish that the transfer operators associated to expanding maps satisfy the assumptions (GL1),...,(GL3). These properties, along with the compact immersion of $W^{k,1}$ in $W^{k-1,1}$ allow one to classically deduce that the transfer operator L_0 of a C^k expanding map T is *quasi-compact* on each $W^{i,1}(S^1)$, with $1 \leqslant i \leqslant k-1$. Furthermore, by topological transitivity of expanding maps, 1 is the only eigenvalue on the unit circle. All of the above leads to the following classical result; see e.g. [25], section 3 or [18] proposition 30. We recall the notation we use for the zero average spaces in $W^{i,1}$

$$V_i := \left\{ g \in W^{i,1}\left(S^1\right) \ s.t. \ \int_{S^1} g \, \mathrm{d}m = 0 \right\} \tag{39}$$

for $i \in \{0, 1, 2\}$.

Proposition 19. *There is $C \geqslant 0$ and $0 < \rho < 1$ such that for each $1 \leqslant i \leqslant k-1$, $g \in V_i$, and $n \geqslant 0$ one has*

$$\|L_0^n g\|_{W^{i,1}} \leqslant C\rho^n \|g\|_{W^{i,1}}.$$

In particular, the resolvent $(Id - L_0)^{-1} := \sum_{j=0}^{\infty} L_0^j$ is a well-defined and bounded operator on V_i.

A.4.2. Small perturbation estimates. In this subsection we recall some more or less known estimates (see e.g. [15]) showing that a small perturbation of an expanding map induces a small perturbation of the associated transfer operator when considered as acting from a stronger to a weaker Sobolev space. This will allow us to verify that the assumption (GL4) applies to suitable deterministic perturbations of expanding maps.

Proposition 20. *Let* $\{T_\delta\}_{\delta\in[0,\bar{\delta})}$ *be a family of* C^2 *expanding maps such that* $T_0 \in C^3$. *Let* L_δ *be the transfer operators associated to* T_δ *and suppose that for some* $K \in \mathbb{R}$ *one has*

$$||T_\delta - T_0||_{C^2} \leqslant K\delta.$$

Then there is a $C > 0$ *such that* $\forall f \in W^{1,1}$:

$$||(L_\delta - L_0)f||_{L^1} \leqslant \delta C||f||_{W^{1,1}} \tag{40}$$

$$||(L_\delta - L_0)f||_{W^{1,1}} \leqslant \delta C||f||_{W^{2,1}}. \tag{41}$$

Proof. In [15], section 7 it is proved that if L_0 and L_δ are transfer operators of C^3 expanding maps T_0 and T_δ, such that for some $K \in \mathbb{R}$

$$||T_\delta - T_0||_{C^2} \leqslant K\delta$$

then there is a $C \in \mathbb{R}$ such that $\forall f \in W^{1,1}$:

$$||(L_\delta - L_0)f||_1 \leqslant \delta C||f||_{W^{1,1}} \tag{42}$$

and (40) is established. From this we can also recover (41), using the explicit formula for the transfer operator:

$$[L_0 f](x) = \sum_{y\in T_0^{-1}(x)} \frac{f(y)}{|T_0'(y)|}. \tag{43}$$

Noting that $T_0'(y) = T_0'(T_0^{-1}(x))$ we can compute the derivative of (43)

$$(L_0 f)' = \sum_{y\in T_0^{-1}(x)} \frac{1}{(T_0'(y))^2} f'(y) - \frac{T_0''(y)}{(T_0'(y))^3} f(y).$$

And similarly for L_δ. Note that

$$(L_0 f)' = L_0\left(\frac{1}{T_0'} f'\right) - L_0\left(\frac{T_0''}{(T_0')^2} f\right). \tag{44}$$

Hence applying (42), (44) and the fact that L is a weak contraction on L^1

$$\begin{aligned}
||(L_\delta - L_0)f||_{W^{1,1}} &\leqslant ||(L_\delta - L_0)f||_{L^1} + ||((L_\delta - L_0)f)'||_{L^1} \\
&\leqslant \delta C||f||_{W^{1,1}} + ||(L_\delta - L_0)\frac{1}{T_0'}f' - (L_\delta - L_0)\frac{T_0''}{(T_0')^2}f||_{L^1} \\
&\quad + ||\frac{1}{T_\delta'}f' - \frac{1}{T_0'}f' + \frac{T_\delta''}{(T_\delta')^2}f - \frac{T_0''}{(T_0')^2}f||_{L^1} \\
&\leqslant \delta C\left(||f||_{W^{1,1}} + ||\frac{1}{T_0'}f'||_{W^{1,1}} + ||\frac{T_0''}{(T_0')^2}f||_{W^{1,1}}\right) \\
&\quad + C_1||T_\delta - T_0||_{C^2}||f||_{W^{1,1}} \\
&\leqslant \delta C_2||f||_{W^{2,1}}
\end{aligned}$$

for some $C_1, C_2 \geqslant 0$ depending on T_0 but not on f. This proves (41). □

Small perturbations in a mixed norm sense—namely $W^{1,1}$ into L^1 – (see (GL4) and proposition 20) will imply a classical fact: the stability of the resolvent. The following will be used in the proof of proposition 3 to verify assumption (27) when invoking theorem 12.

Proposition 21. *For $\delta \in [0,\overline{\delta})$, let $T_\delta : S^1 \to S^1$ be a family of C^3 expanding maps. Suppose that the dependence of the family on δ is differentiable at 0 in the sense of assumptions (A0), (A1),(A2), then*

$$\lim_{\delta\to 0} ||\,(Id - L_\delta)^{-1} - (Id - L_0)^{-1}\,||_{V_1 \to V_0} = 0 \tag{45}$$

where the spaces V_i are defined as in (39).

Proof. The result follows from Theorem 1 of [21]. This theorem says that if a family of bounded linear operators L_δ acting on weak and strong Banach spaces $\mathcal{B}_s, \mathcal{B}_w$ such that (i) there is a compact immersion of $\mathcal{B}_s$ into $\mathcal{B}_w$, (ii) the family of operators satisfy (GL1), the first equation of (GL2), (GL3) and (GL4), and (iii) the spectral radius of $L_0 : \mathcal{B}_s \to \mathcal{B}_s$ is strictly smaller than 1,

$$\lim_{\delta\to 0} ||\,(Id - L_\delta)^{-1} - (Id - L_0)^{-1}\,||_{\mathcal{B}_s \to \mathcal{B}_w} = 0.$$

Equation (45) will follow directly by applying this theorem to the transfer operators L_δ associated to a family of expanding maps T_δ satisfying (A0), (A1), (A2) considering $\mathcal{B}_s = V_1$ and $\mathcal{B}_w = V_0$. For (i) the compact immersion is well known from the Rellich–Kondracov theorem. For (ii), note that for the family of maps T_δ, (GL1), (GL2) and (GL3) are verified by the Lasota–Yorke inequalities established in lemma 18, and the small perturbation assumptions needed at (GL4) are established in Proposition 20. For (iii), we note that by Proposition 19, the spectral radius of $L_0|_{V_1} \leqslant \rho < 1$. Thus, noting that $\|(Id - L_0)^{-1}\|_{V_1} < \infty$ (also by Proposition 19) the theorem can be applied, implying (45). □

A.4.3. The derivative operator. In this section we study the operator $\dot{L}$ representing a 'first derivative' of the transfer operator with respect to the perturbation parameter δ. The next result is similar to [17, Proposition 3.1] but has been strengthened to be quantitative and uniform, in order to verify the assumptions (GL5), (GL6) of theorem 13 and the assumption (25) of proposition 16.

Proposition 22. *Let $T_\delta : S^1 \to S^1$, where $\delta \in [0,\overline{\delta})$ be a family of C^3 expanding maps. Suppose that the dependence of the family on δ is differentiable at 0 in the sense of (A2). Let L_δ be the transfer operator associated to T_δ. Let us consider again the operator $\dot{L} : W^{1,1} \to L^1$ defined in (3) by*

$$\dot{L}(w) = -L_0\left(\frac{w\dot{T}'}{T_0'}\right) - L_0\left(\frac{\dot{T}w'}{T_0'}\right) + L_0\left(\frac{\dot{T}T_0''}{T_0'^2}w\right). \tag{46}$$

For $w \in C^2(S^1, \mathbb{R})$ we have

$$\dot{L}(w) = \lim_{\delta\to 0}\left(\frac{L_\delta w - L_0 w}{\delta}\right) \tag{47}$$

with convergence in the C^1 topology (and therefore also in the $W^{1,1}$ and H^1 topologies). Moreover, one has a quantitative and uniform convergence on the unit ball $B_{W^{2,1}(S^1)}$:

$$\sup_{w\in B_{W^{2,1}(S^1)}}\left\|\dot{L}(w)-\frac{(L_\delta-L_0)w}{\delta}\right\|_{L^1}=O(\delta). \tag{48}$$

We denote by $\{y_i^\delta\}_{i=1}^d := T_\delta^{-1}(x)$ and $\{y_i^0\}_{i=1}^d := T_0^{-1}(x)$ the d preimages under T_δ and T_0, respectively, of a point $x\in S^1$. Futhermore, we assume that the indexing is chosen so that y_i^δ is a small perturbation of y_i^0, for $1\leqslant i\leqslant d$.

The proof of proposition 22 will be given at the end of this section, after the proof of two preliminary Lemma.

Lemma 19. *For $y_i^\delta\in T_\delta^{-1}(x)$ we can write*

$$y_i^\delta=y_i^0+\delta\left(-\frac{\dot{T}(y_i^0)}{T_0'(y_i^0)}\right)+O_{C^1}(\delta^2),\quad \textit{in the limit as } \delta\to 0,$$

where we say F_δ is $O_{C^1}(\delta^2)$ if

$$\sup_{\delta\in(0,\overline{\delta})}\frac{\|F_\delta\|_{C^1}}{\delta^2}<+\infty. \tag{49}$$

Proof of lemma 19. For a family of functions F_δ we will say that $F_\delta=O_{C^0}(\delta^2)$ if $\sup_{\delta\in(0,\overline{\delta})}\frac{||F_\delta||_{C^0}}{\delta^2}<+\infty$ for some $\overline{\delta}>0$. Let $T_{\delta,i}$ be the branches of T_δ. We have $y_i^\delta(x)=T_{\delta,i}^{-1}(x)$ and

$$\frac{\mathrm{d}y_i^\delta(x)}{\mathrm{d}x}=\frac{1}{T_\delta'\left(T_{\delta,i}^{-1}(x)\right)}. \tag{50}$$

Now let us compute $\frac{\mathrm{d}y_i^\delta(x)}{\mathrm{d}\delta}$. Let us fix $x\in S^1$ and write

$$y_i^\delta(x)=y_i^0(x)+\delta\epsilon_i(x)+F_i(\delta,x) \tag{51}$$

where for each x, $F_i(\delta,x)=o(\delta)$ as $\delta\to 0$. We will show that $\epsilon_i(x)=-\frac{\dot{T}(y_i^0(x))}{T_0'(y_i^0(x))}$ and $F_i(\delta,x)=O(\delta^2)$ for each x, then we will show that $F_i(\delta,x)$ satisfies (49).

For the first two claims, let us fix $x\in S^1$. Substituting (51) into the identity $T_\delta(y_i^\delta(x))=x$ we can expand

$$\begin{aligned}x&=T_\delta\left(y_i^\delta(x)\right)\\&=T_0\left(y_i^\delta(x)\right)+\delta\dot{T}\left(y_i^\delta(x)\right)+E\left(\delta,y_i^\delta(x)\right)\end{aligned}$$

where by (2), $E(\delta,x)$ satisfies (49). Further,

$$\begin{aligned}x=&\,T_0\left(y_i^0(x)+\delta\epsilon_i(x)+F_i(\delta,x)\right)\\&+\delta\dot{T}\left(y_i^0(x)+\delta\epsilon_i(x)+F_i(\delta,x)\right)+E\left(\delta,y_i^\delta(x)\right).\end{aligned} \tag{52}$$

Since $T_0 \in C^2$ we can write the first term in the right hand side of (52) as

$$T_0\left(y_i^0(x) + \delta\epsilon_i(x) + F_i(\delta,x)\right) = T_0\left(y_i^0(x)\right) + T_0'\left(y_i^0(x)\right)\left(\delta\epsilon_i(x) + F_i(\delta,x)\right) + \frac{1}{2}T_0^{''}(\xi)\left(\delta\epsilon_i(x) + F_i(\delta,x)\right)^2$$

for some $\xi \in S^1$. Since $\dot{T} \in C^2$ we can develop the second term in (52) as

$$\delta\dot{T}\left(y_i^0(x) + \delta\epsilon_i(x) + F_i(\delta,x)\right) = \delta\dot{T}\left(y_i^0(x)\right) + \delta\dot{T}'\left(y_i^0(x)\right)\left(\delta\epsilon_i(x) + F_i(\delta,x)\right) + \frac{1}{2}\dot{T}_0^{''}(\xi')\left(\delta\epsilon_i(x) + F_i(\delta,x)\right)^2$$

for some $\xi' \in S^1$ and use that $T_0(y_i^0(x)) = x$ to cancel terms on either side of (52) and get

$$\begin{aligned}0 &= T_0'\left(y_i^0(x)\right)\left(\delta\epsilon_i(x) + F_i(\delta,x)\right) + \delta\dot{T}\left(y_i^0(x)\right) + \delta\dot{T}'\left(y_i^0(x)\right)\left(\delta\epsilon_i(x) + F_i(\delta,x)\right) \\ &\quad + \frac{1}{2}T_0^{''}(\xi')\left(\delta\epsilon_i(x) + F_i(\delta,x)\right)^2 + \frac{1}{2}\dot{T}_0^{''}(\xi')\left(\delta\epsilon_i(x) + F_i(\delta,x)\right)^2 + E\left(\delta, y_i^\delta(x)\right). \qquad (53)\end{aligned}$$

Recalling that $T_0^{''}$ is uniformly bounded on S^1, for each fixed x, as $\delta \to 0$ we can then identify the first-order terms in (53) as $\delta T_0'(y_i^0(x))\epsilon_i(x) + \delta\dot{T}(y_i^0(x))$, giving $\epsilon_i(x) = -\frac{\dot{T}(y_i^0)}{T_0'(y_i^0)}$ and thus

$$\frac{dy_i^\delta(x)}{d\delta} = -\frac{\dot{T}\left(y_i^0(x)\right)}{T_0'\left(y_i^0(x)\right)}.$$

Putting $\epsilon_i(x) = -\frac{\dot{T}\circ y_i^0}{T_0'\circ y_i^0}$ into (53) we get

$$\begin{aligned}0 &= T_0'\left(y_i^0(x)\right)F_i(\delta,x) + \delta\dot{T}'\left(y_i^0(x)\right)F_i(\delta,x \\ &\quad + \frac{1}{2}T_0^{''}(\xi')\left(\delta\epsilon_i(x) + F_i(\delta,x)\right)^2 + \frac{1}{2}\dot{T}_0^{''}(\xi')\left(\delta\epsilon_i(x) + F_i(\delta,x)\right)^2 + E\left(\delta, y_i^\delta(x)\right)\end{aligned}$$

from which

$$|T_0'\left(y_i^0(x)\right)F_i(\delta,x) + \delta\dot{T}'\left(y_i^0(x)\right)F_i(\delta,x)| \leqslant |E\left(\delta, y_i^\delta(x)\right)| + M\left(\delta\epsilon_i(x) + F_i(\delta,x)\right)^2$$

where $M = \sup_{x\in S^1}|\max(T_0^{''}(x), \dot{T}_0^{''}(x))|$. Considering δ small enough we can suppose $T_0'(y_i^0(x)) + \delta\dot{T}'(y_i^0(x)) \geqslant 1$, hence

$$|F_i(\delta,x)| \leqslant |E\left(\delta, y_i^\delta(x)\right)| + M\left(\delta^2\epsilon_i(x) + F_i(\delta,x)^2 + 2\delta\epsilon_i(x)F_i(\delta,x)\right). \qquad (54)$$

Since $F_i(\delta,x) = o(\delta)$ we get that for each x

$$F_i(\delta,x) = O\left(\delta^2\right). \qquad (55)$$

By (51) we have

$$F_i(\delta,x) = y_i^\delta(x) - y_i^0(x) + \delta\frac{\dot{T}\left(y_i^0(x)\right)}{T_0'\left(y_i^0(x)\right)}$$

and by (50)

$$\frac{\mathrm{d}F_i(\delta,x)}{\mathrm{d}x} = \frac{1}{T_\delta'\left(y_i^\delta(x)\right)} - \frac{1}{T_0'\left(y_i^0(x)\right)} + \delta\frac{y_i^0(x)'\dot{T}'\left(y_i^0(x)\right)T_0'\left(y_i^0(x)\right) - y_i^0(x)'\dot{T}\left(y_i^0(x)\right)T_0''\left(y_i^0(x)\right)}{\left(T_0'\left(y_i^0(x)\right)\right)^2}. \tag{56}$$

This shows that $\frac{\mathrm{d}F_i(\delta,x)}{\mathrm{d}x}$ is uniformly bounded and then by (55) and the compactness of S^1

$$\lim_{\delta\to 0}||F_i(\delta,x)||_\infty = 0$$

for each i. This means that for δ small enough $|F_i(\delta,x)| \leqslant \frac{1}{2M}$ for each x and then inserting this in (54) we get

$$|F_i(\delta,x)| \leqslant |E\left(\delta,y_i^\delta(x)\right)| + M\left(\delta^2\epsilon_i(x) + \frac{1}{2M}|F_i(\delta,x)| + 2\delta\epsilon_i(x)|F_i(\delta,x)|\right)$$

and

$$\begin{aligned}|F_i(\delta,x)| &\leqslant 2|E\left(\delta,y_i^\delta(x)\right)| + 2M\left(\delta^2\epsilon_i(x) + 2\delta\epsilon_i(x)|F_i(\delta,x)|\right)\\ (1-4M\delta\epsilon_i(x))|F_i(\delta,x)| &\leqslant 2|E\left(\delta,y_i^\delta(x)\right)| + 2M\left(\delta^2\epsilon_i(x)\right)\\ |F_i(\delta,x)| &\leqslant \frac{2|E\left(\delta,y_i^\delta(x)\right)| + 2M\left(\delta^2\epsilon_i(x)\right)}{(1-4M\delta\epsilon_i(x))}\end{aligned}$$

from which we get that

$$F_i(\delta,x) = O_{C^0}\left(\delta^2\right).$$

Now we prove that

$$\frac{\mathrm{d}F_i(\delta,x)}{\mathrm{d}x} = O_{C^0}\left(\delta^2\right).$$

Recalling (56) we have

$$\frac{\mathrm{d}F_i(\delta,x)}{\mathrm{d}x} = \frac{1}{T_\delta'\left(y_i^\delta(x)\right)} - \frac{1}{T_0'\left(y_i^0(x)\right)} + \delta\frac{y_i^0(x)'\dot{T}'\left(y_i^0(x)\right)T_0'\left(y_i^0(x)\right) - y_i^0(x)'\dot{T}\left(y_i^0(x)\right)T_0''\left(y_i^0(x)\right)}{\left(T_0'\left(y_i^0(x)\right)\right)^2}.$$

Now we analyze the fist part of the right hand side

$$\frac{1}{T_\delta'\left(y_i^\delta(x)\right)} - \frac{1}{T_0'\left(y_i^0(x)\right)} = \left[\frac{1}{T_\delta'\left(y_i^\delta(x)\right)} - \frac{1}{T_\delta'\left(y_i^0(x)\right)}\right] + \left[\frac{1}{T_\delta'\left(y_i^0(x)\right)} - \frac{1}{T_0'\left(y_i^0(x)\right)}\right] \tag{57}$$

where for the first summand, since $y_i^\delta(x) = y_i^0(x) + \delta\epsilon_i(x) + F_i(\delta,x)$ we get

$$\begin{aligned}
\left[\frac{1}{T'_\delta\left(y_i^\delta(x)\right)} - \frac{1}{T'_\delta\left(y_i^0(x)\right)}\right] &= \left[\frac{1}{T'_\delta\left(y_i^0(x) + \delta\epsilon_i(x) + F_i(\delta,x)\right)} - \frac{1}{T'_\delta\left(y_i^0(x)\right)}\right] \\
&= \frac{-T''_\delta\left(y_i^0(x)\right)}{T'_\delta\left(y_i^0(x)\right)^2}\left[\delta\epsilon_i(x) + F_i(\delta,x)\right] \\
&\quad + \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{-T''_\delta}{T'^2_\delta}\right](\xi)\,\frac{1}{2}\left[\delta\epsilon_i(x) + F_i(\delta,x)\right]^2
\end{aligned}$$

for some $\xi \in S^1$ (depending on x), by Lagrange theorem. Then we get

$$\begin{aligned}
\left[\frac{1}{T'_\delta\left(y_i^\delta(x)\right)} - \frac{1}{T'_\delta\left(y_i^0(x)\right)}\right] &= \frac{-T''_\delta\left(y_i^0(x)\right)}{T'_\delta\left(y_i^0(x)\right)^2}\left[\delta\frac{-\dot{T}\left(y_i^0(x)\right)}{T'_0\left(y_i^0(x)\right)} + F_i(\delta,x)\right] \\
&\quad + \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{-T''_\delta}{T'^2_\delta}\right](\xi)\left[\delta\epsilon_i(x) + F_i(\delta,x)\right]^2 \\
&= \frac{-[T''_0\left(y_i^0(x)\right) + \delta\dot{T}''\left(y_i^0(x)\right) + E''\left(\delta,y_i^0(x)\right)]}{[T'_0\left(y_i^0(x)\right) + \delta\dot{T}'(y_i^0(x)) + E'(\delta,y_i^0(x))]^2}\left[\delta\frac{-\dot{T}(y_i^0(x))}{T'_0(y_i^0(x))} + F_i(\delta,x)\right] \\
&\quad + \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{-T''_\delta}{T'^2_\delta}\right](\xi)\,\frac{1}{2}([\delta\epsilon_i(x) + F_i(\delta,x)]^2).
\end{aligned}$$

Since $\frac{\mathrm{d}}{\mathrm{d}x}[\frac{-T''_\delta}{T'^2_\delta}]$ is bounded we get

$$\begin{aligned}
\left[\frac{1}{T'_\delta\left(y_i^\delta(x)\right)} - \frac{1}{T'_\delta\left(y_i^0(x)\right)}\right] &= \frac{\delta\dot{T}\left(y_i^0(x)\right)[T''_0\left(y_i^0(x)\right) + \delta\dot{T}''\left(y_i^0(x)\right) + E''\left(\delta,y_i^0(x)\right)]}{T'_0\left(y_i^0(x)\right)[T'_0\left(y_i^0(x)\right) + \delta\dot{T}'\left(y_i^0(x)\right) + E'\left(\delta,y_i^0(x)\right)]^2} + O_{C^0}\left(\delta^2\right) \\
&= \frac{\delta T''_0\left(y_i^0(x)\right)\dot{T}\left(y_i^0(x)\right)}{[T'_0(y_i^0(x))]^3 + 2\delta\dot{T}'(y_i^0(x))T'_0(y_i^0(x)) + O_{C^0}(\delta^2)} + O_{C^0}(\delta^2) \\
&= \frac{\delta T''_0(y_i^0(x))\dot{T}(y_i^0(x))}{[T'_0(y_i^0(x))]^3} + O_{C^0}(\delta^2).
\end{aligned}$$

The second summand is expanded similarly as follows

$$\begin{aligned}
\left[\frac{1}{T'_\delta\left(y_i^0(x)\right)} - \frac{1}{T'_0\left(y_i^0(x)\right)}\right] &= \left[\frac{1}{T'_0\left(y_i^0(x)\right) + \delta\dot{T}'\left(y_i^0(x)\right) + E'\left(\delta,y_i^0(x)\right)} - \frac{1}{T'_0\left(y_i^0(x)\right)}\right] \\
&= \frac{-1}{T'_0\left(y_i^0(x)\right)^2}\left[\delta\dot{T}'\left(y_i^0(x)\right) + E'\left(\delta,y_i^0(x)\right)\right] \\
&\quad + \frac{1}{2}\frac{2}{(\xi)^3}\left[\delta\dot{T}'\left(y_i^0(x)\right) + E'\left(\delta,y_i^0(x)\right)\right]^2 \\
&= \frac{-\delta\dot{T}'\left(y_i^0(x)\right)T'_0\left(y_i^0(x)\right)}{T'_0\left(y_i^0(x)\right)^3} + O_{C^0}\left(\delta^2\right).
\end{aligned} \tag{58}$$

since applying the Lagrange theorem $|\xi - T'_0(y_i^0(x))|$ can be small as wanted when δ is small enough and then ξ is bounded away from 0.

Putting the two summands expanded together and recalling that $\frac{\mathrm{d}y_i^\delta(x)}{\mathrm{d}x} = \frac{1}{T_\delta'(T_{\delta,i}^{-1}(x))} = \frac{1}{T_\delta'(y_i^\delta(x))}$ we get

$$\begin{aligned}
\frac{\mathrm{d}F_i(\delta,x)}{\mathrm{d}x} &= \frac{\delta T_0''\left(y_i^0(x)\right)\dot{T}\left(y_i^0(x)\right)}{\left[T_0'\left(y_i^0(x)\right)\right]^3} - \frac{\delta\dot{T}'\left(y_i^0(x)\right)T_0'\left(y_i^0(x)\right)}{T_0'\left(y_i^0(x)\right)^3} \\
&\quad + \delta\frac{d}{dx}\left[\frac{\dot{T}\left(y_i^0(x)\right)}{T_0'\left(y_i^0(x)\right)}\right] + O_{C^0}\left(\delta^2\right) \\
&= \frac{\delta T_0''\left(y_i^0(x)\right)\dot{T}\left(y_i^0(x)\right)}{\left[T_0'\left(y_i^0(x)\right)\right]^3} - \frac{\delta\dot{T}'\left(y_i^0(x)\right)T_0'\left(y_i^0(x)\right)}{T_0'\left(y_i^0(x)\right)^3} \\
&\quad + \delta\frac{y_i^0(x)'\dot{T}'\left(y_i^0(x)\right)T_0'\left(y_i^0(x)\right) - y_i^0(x)'T_0''\left(y_i^0(x)\right)\dot{T}\left(y_i^0(x)\right)}{\left(T_0'\left(y_i^0(x)\right)\right)^2} + O_{C^0}\left(\delta^2\right) \\
&= O_{C^0}\left(\delta^2\right)
\end{aligned}$$

proving the statement. □

The following simple lemma on the convergence of compositions of functions will also be useful.

Lemma 24. *Let $\overline{\delta} > 0, K_0 > 0, K_1 > 0$. Suppose the following hold when $\delta \in [0,\overline{\delta})$:*
$f_\delta, g_\delta \in C^1(S^1,\mathbb{R})$, $||f_\delta||_{C^1} \leqslant K_0$, $||g_\delta||_{C^1} \leqslant K_0$, $||f_\delta - f_0||_{C^1} \to 0$ *and* $||g_\delta - g_0||_{C^1} \to 0$.
Then

$$\lim_{\delta\to 0}||f_\delta\circ g_\delta - f_0\circ g_0||_{C^1} = 0. \tag{59}$$

If, in addition, $||f_\delta - f_0||_{C^0} \leqslant K_1\delta$, and $||g_\delta - g_0||_{C^0} \leqslant K_1\delta$ then there is $K_2 \geqslant 0$ such that for $\delta \in [0,\eta]$

$$||f_\delta\circ g_\delta - f_0\circ g_0||_{C^0} \leqslant K_2\delta \tag{60}$$

with K_2 only depending on $Lip(f_0)$ and K_1.

Proof. To prove (59) we can write

$$\begin{aligned}
||\left(f_\delta\circ g_\delta - f_0\circ g_0\right)'||_\infty &= ||\left(f_\delta\circ g_\delta - f_0\circ g_\delta + f_0\circ g_\delta - f_0\circ g_0\right)'||_\infty \\
&\leqslant ||\left(f_\delta\circ g_\delta - f_0\circ g_\delta\right)'||_\infty + ||\left(f_0\circ g_\delta - f_0\circ g_0\right)'||_\infty
\end{aligned}$$

and while it is obvious that the first summand goes to 0, for the second we have

$$\begin{aligned}
||\left(f_0\circ g_\delta - f_0\circ g_0\right)'||_\infty &= ||g_\delta' f_0'\circ g_\delta - g_0' f_0'\circ g_0||_\infty \\
&\leqslant ||g_\delta' f_0'\circ g_\delta - g_\delta' f_0'\circ g_0||_\infty + ||g_\delta' f_0'\circ g_0 - g_0' f_0'\circ g_0||_\infty
\end{aligned}$$

where $||g_\delta' f_0'\circ g_\delta - g_\delta' f_0'\circ g_0||_\infty \to 0$ because of the uniform continuity of f_0' and the uniform bound on $||g_\delta||_{C^1}$ while $||g_\delta' f_0'\circ g_0 - g_0' f_0'\circ g_0||_\infty \to 0$ since $g_\delta \to g_0$ in C^1, and (59) is proved.

To prove (60) we write

$$\begin{aligned}
||f_\delta\circ g_\delta - f_0\circ g_0||_\infty &= ||f_\delta\circ g_\delta - f_0\circ g_\delta + f_0\circ g_\delta - f_0\circ g_0||_\infty \\
&= ||f_\delta\circ g_\delta - f_0\circ g_\delta||_\infty + ||f_0\circ g_\delta - f_0\circ g_0||_\infty
\end{aligned}$$

and note that

$$||f_0 \circ g_\delta - f_0 \circ g_0||_\infty \leqslant Lip(f_0) K_1 \delta.$$

while

$$||f_\delta \circ g_\delta - f_0 \circ g_\delta||_\infty \leqslant K_1 \delta$$

and (60) is proved. □

We now prove proposition 22.

Proof of proposition 22 . First we introduce a notation we will use in the proof to simplify some equation. For $k \geqslant 0$ we say that a certain family of functions $F_\delta : S^1 \to \mathbb{R}$ is $O_{C^k}(\delta^2)$ if $\exists \overline{\delta} > 0$ such that $\sup_{\delta \in (0,\overline{\delta})} \frac{\|F_\delta\|_{C^k}}{\delta^2} < +\infty$. We again denote by $\{y_i^\delta\}_{i=1}^d := T_\delta^{-1}(x)$ and $\{y_i^0\}_{i=1}^d := T_0^{-1}(x)$ the d preimages under T_δ and T_0, respectively, of a point $x \in X$. We can write

$$\begin{aligned}
\frac{L_\delta w(x) - L_0 w(x)}{\delta} &= \frac{1}{\delta}\left(\sum_{i=1}^d \frac{w\left(y_i^\delta\right)}{T_\delta'\left(y_i^\delta\right)} - \sum_{i=1}^d \frac{w\left(y_i^0\right)}{T_0'\left(y_i^0\right)}\right) \\
&= \underbrace{\frac{1}{\delta}\left(\sum_{i=1}^d w\left(y_i^\delta\right)\left(\frac{1}{T_\delta'\left(y_i^\delta\right)} - \frac{1}{T_0'\left(y_i^\delta\right)}\right)\right)}_{=:(I)} + \underbrace{\frac{1}{\delta}\left(\sum_{i=1}^d \frac{w\left(y_i^\delta\right) - w\left(y_i^0\right)}{T_0'\left(y_i^\delta\right)}\right)}_{=:(II)} \\
&\quad + \underbrace{\frac{1}{\delta}\left(\sum_{i=1}^d w\left(y_i^0\right)\left(\frac{1}{T_0'\left(y_i^\delta\right)} - \frac{1}{T_0'\left(y_i^0\right)}\right)\right)}_{=:(III)}. \qquad (61)
\end{aligned}$$

In the following we will analyze the terms $(I), (II), (III)$ showing the convergence of these terms to the three summands in the right hand side of (46) both in the C^1 topology (see (47)) and in the uniform sense required by (48).

Term (I): For the first term we first differentiate the expansion $T_\delta(x) = T_0(x) + \delta\dot{T}(x) + O_{C^2}(\delta^2)$ in x to get:

$$T_\delta'(x) = T_0'(x) + \delta\dot{T}'(x) + O_{C^1}\left(\delta^2\right). \qquad (62)$$

We can then write

$$\begin{aligned}
(I) &= \frac{1}{\delta}\left(\sum_{i=1}^d w\left(y_i^\delta\right)\left(\frac{1}{T_\delta'\left(y_i^\delta\right)} - \frac{1}{T_0'\left(y_i^\delta\right)}\right)\right) \\
&= \frac{1}{\delta}\left(\sum_{i=1}^d \frac{w\left(y_i^\delta\right)}{T_\delta'\left(y_i^\delta\right)}\left(1 - \frac{T_\delta'\left(y_i^\delta\right)}{T_0'\left(y_i^\delta\right)}\right)\right) \\
&= \frac{1}{\delta}\left(\sum_{i=1}^d \frac{w\left(y_i^\delta\right)}{T_\delta'\left(y_i^\delta\right)}\left(1 - \left(\frac{T_0'\left(y_i^\delta\right) + \delta\dot{T}'\left(y_i^\delta\right) + O_{C^1}\left(\delta^2\right)}{T_0'\left(y_i^\delta\right)}\right)\right)\right) \\
&= \left(-\sum_{i=1}^d \frac{w\left(y_i^\delta\right)\dot{T}'\left(y_i^\delta\right)}{T_\delta'\left(y_i^\delta\right)T_0'\left(y_i^\delta\right)}\right) + \frac{O_{C^1}\left(\delta^2\right)}{\delta}. \qquad (63)
\end{aligned}$$

Since $w(\cdot)\dot{T}'(\cdot)/(T'_\delta(\cdot)T'_0(\cdot))$ is C^1 on the circle with uniformly bounded norm when δ is small enough, we have that

$$\begin{aligned}\lim_{\delta\to 0}\frac{1}{\delta}\left(\sum_{i=1}^{d} w\left(y_i^\delta(\cdot)\right)\left(\frac{1}{T'_\delta\left(y_i^\delta(\cdot)\right)}-\frac{1}{T'_0\left(y_i^\delta(\cdot)\right)}\right)\right) &= \lim_{\delta\to 0}\left(-\sum_{i=1}^{d}\frac{w\left(y_i^\delta(\cdot)\right)\dot{T}'\left(y_i^\delta(\cdot)\right)}{T'_\delta\left(y_i^\delta(\cdot)\right)T'_0\left(y_i^\delta(\cdot)\right)}\right)\\ &= -L_0\left(\frac{w\dot{T}'}{T'_0}\right), \end{aligned} \tag{64}$$

with convergence in the C^1 topology, using lemma 19, (59), and (62) in the final equality. This proves that (I) converges in the C^1 topology to the first summand of (46).

We now turn to the uniform convergence in (48) for term (I). By uniform expansivity of T_0 and (62) there is K_1 such that

$$\left\|\frac{w(\cdot)\dot{T}'(\cdot)}{T'_\delta(\cdot)T'_0(\cdot)}-\frac{w(\cdot)\dot{T}'(\cdot)}{(T'_0(\cdot))^2}\right\|_{C^0}\leqslant\left\|\frac{1}{T'_\delta(\cdot)}-\frac{1}{T'_0(\cdot)}\right\|_{C^0}\left\|\frac{w(\cdot)\dot{T}'(\cdot)}{T'_0(\cdot)}\right\|_{C^0}\leqslant K_1\delta \tag{65}$$

when δ is sufficiently small. By the Sobolev embedding theorem, K_1 is uniform for $w\in B_{W^{2,1}(0,1)}$, (the unit ball of $W^{2,1}$). We apply (65) and (60) to $L_\delta(w\dot{T}'/T'_0)=\left(-\sum_{i=1}^{d}\frac{w(y_i^\delta)\dot{T}'(y_i^\delta)}{T'_\delta(y_i^\delta)T'_0(y_i^\delta)}\right)$ and $L_0(w\dot{T}'/T'_0)=\left(-\sum_{i=1}^{d}\frac{w(y_i^0)\dot{T}'(y_i^0)}{T'_0(y_i^0)T'_0(y_i^0)}\right)$ to obtain

$$\left\|L_\delta\left(w\dot{T}'/T'_0\right)-L_0\left(w\dot{T}'/T'_0\right)\right\|_{C^0}=O(\delta). \tag{66}$$

Note that the constant K_2 in (60) only depends on $\mathrm{Lip}(f_0)$ and K_1.

Substituting $L_\delta(w\dot{T}'/T'_0)=\left(-\sum_{i=1}^{d}\frac{w(y_i^\delta)\dot{T}'(y_i^\delta)}{T'_\delta(y_i^\delta)T'_0(y_i^\delta)}\right)$, into (63), we see that as $\delta\to 0$ one has

$$\frac{1}{\delta}\left(\sum_{i=1}^{d} w\left(y_i^\delta\right)\left(\frac{1}{T'_\delta\left(y_i^\delta\right)}-\frac{1}{T'_0\left(y_i^\delta\right)}\right)\right)=L_\delta\left(w\dot{T}'/T'_0\right)+\frac{O_{C^1}\left(\delta^2\right)}{\delta}.$$

By (66), we obtain

$$\left\|\frac{1}{\delta}\left(\sum_{i=1}^{d} w\left(y_i^\delta\right)\left(\frac{1}{T'_\delta\left(y_i^\delta\right)}-\frac{1}{T'_0\left(y_i^\delta\right)}\right)\right)+L_0\left(\frac{w\dot{T}'}{T'_0}\right)\right\|_{C^0}=O(\delta),$$

and again by the Sobolev embedding theorem, and the uniformity of constants noted immediately after (66) this is uniform for $w\in B_{W^{2,1}}(0,1)$.

Term (II): we prove the convergence of the second term of (61) to the second summand of (46) both in the sense of (47) and (48). Suppose $w\in C^2$. Using the Taylor expansion with Lagrange remainder we have that

$$\begin{aligned} w\left(y_i^\delta\right) &= w\left(y_i^0+y_i^\delta-y_i^0\right)\\ &= w\left(y_i^0\right)+w'\left(y_i^0\right)[y_i^\delta-y_i^0]+\frac{1}{2}w''(\xi)\left([y_i^\delta-y_i^0]\right)^2, \end{aligned}$$

where ξ lies between y_i^δ and y_i^0.

Using lemma 19 we get

$$w(y_i^\delta) = w(y_i^0) + w'(y_i^0)\left[-\delta\frac{\dot T(y_i^0)}{T_0'(y_i^0)} + O_{C^0}(\delta^2)\right] + \frac{1}{2}w''(\xi)\left(\delta\left(-\frac{\dot T(y_i^0)}{T_0'(y_i^0)}\right) + O_{C^0}(\delta^2)\right)^2.$$

Since w'' is uniformly bounded, we get

$$w\left(y_i^\delta\right) = w\left(y_i^0\right) + w'\left(y_i^0\right) - \delta\left[\frac{\dot T\left(y_i^0\right)}{T_0'\left(y_i^0\right)}\right] + O_{C^0}\left(\delta^2\right).$$

Thus,

$$\begin{aligned}(II) = \frac{1}{\delta}\sum_{i=1}^{d}\frac{w\left(y_i^\delta\right) - w\left(y_i^0\right)}{T_0'\left(y_i^\delta\right)} &= -\frac{1}{\delta}\sum_{i=1}^{d}\frac{\delta w'\left(y_i^0\right)\frac{\dot T(y_i^0)}{T_0'(y_i^0)} + O_{C^0}\left(\delta^2\right)}{T_0'\left(y_i^\delta\right)} \\ &= -\sum_{i=1}^{d}\frac{\dot T\left(y_i^0\right)w'\left(y_i^0\right)}{T_0'\left(y_i^0\right)T_0'\left(y_i^\delta\right)} + \frac{O_{C^0}\left(\delta^2\right)}{\delta}.\end{aligned}$$

As in (64), by lemma 19, one has in the C^0 topology

$$\lim_{\delta\to 0}\frac{1}{\delta}\sum_{i=1}^{d}\frac{w\left(y_i^\delta\right) - w\left(y_i^0\right)}{T_0'\left(y_i^\delta\right)} = -L_0\left(\frac{\dot T w'}{T_0'}\right)(x). \tag{67}$$

Now we prove the convergence in C^1. To begin this we first prove that in the C^1 topology

$$\lim_{\delta\to 0}\frac{1}{\delta}\sum_{i=1}^{d}\frac{w\left(y_i^\delta\right) - w\left(y_i^0\right)}{T_0'\left(y_i^\delta\right)} = \lim_{\delta\to 0}\frac{1}{\delta}\sum_{i=1}^{d}\frac{w\left(y_i^\delta\right) - w\left(y_i^0\right)}{T_0'\left(y_i^0\right)}. \tag{68}$$

We note that

$$\begin{aligned}&\frac{1}{\delta}\sum_{i=1}^{d}\left[\frac{w\left(y_i^\delta\right) - w\left(y_i^0\right)}{T_0'\left(y_i^\delta\right)} - \frac{w\left(y_i^\delta\right) - w\left(y_i^0\right)}{T_0'\left(y_i^0\right)}\right] \\ &\quad = \frac{1}{\delta}\sum_{i=1}^{d}\frac{\left[-T_0'\left(y_i^\delta\right) + T_0'\left(y_i^0\right)\right]\left[w\left(y_i^\delta\right) - w\left(y_i^0\right)\right]}{T_0'\left(y_i^\delta\right)T_0'\left(y_i^0\right)} \\ &\quad = \frac{1}{\delta}\sum_{i=1}^{d}\frac{\left[\delta T_0''\left(y_i^0\right)\frac{\dot T(y_i^0)}{T_0'(y_i^0)} + O_{C^1}\left(\delta^2\right)\right]\left[w\left(y_i^\delta\right) - w\left(y_i^0\right)\right]}{T_0'\left(y_i^\delta\right)T_0'\left(y_i^0\right)},\end{aligned}$$

using lemma 19 for the second equality. Since $w(y_i^\delta) \to w(y_i^0)$ in C^1 then by the expression immediately above, $\frac{1}{\delta}\sum_{i=1}^{d}[\frac{w(y_i^\delta)-w(y_i^0)}{T_0'(y_i^\delta)} - \frac{w(y_i^\delta)-w(y_i^0)}{T_0'(y_i^0)}]$ tends to 0 in C^1 and (68) is proved. Using this we now upgrade the convergence of (67) to C^1. By (68) it is sufficient to prove

$$\lim_{\delta\to 0}\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{w\left(y_i^\delta\right) - w\left(y_i^0\right)}{\delta T_0'\left(y_i^0\right)}\right] = -\frac{\mathrm{d}}{\mathrm{d}x}\frac{\dot T\left(y_i^0\right)w'\left(y_i^0\right)}{T_0'\left(y_i^0\right)^2}$$

in the C^0 topology. By uniform expansivity of T_0 this is equivalent to proving

$$\lim_{\delta\to 0}\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{w\left(y_i^\delta\right)-w\left(y_i^0\right)}{\delta}\right]$$
$$=-\frac{\mathrm{d}}{\mathrm{d}x}\frac{\dot{T}\left(y_i^0\right)w'\left(y_i^0\right)}{T_0'\left(y_i^0\right)}$$
$$=-\frac{1}{T_0'\left(y_i^0\right)}\frac{\left[\dot{T}'\left(y_i^0\right)w'\left(y_i^0\right)+w''\left(y_i^0\right)\dot{T}\left(y_i^0\right)\right]T_0'\left(y_i^0\right)-\dot{T}\left(y_i^0\right)w'\left(y_i^0\right)T_0''\left(y_i^0\right)}{T_0'\left(y_i^0\right)^2} \tag{69}$$

in the C^0 topology. Thus let us compute

$$\lim_{\delta\to 0}\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{w\left(y_i^\delta\right)-w\left(y_i^0\right)}{\delta}\right]=\lim_{\delta\to 0}\frac{\frac{1}{T_\delta'\left(y_i^\delta\right)}w'\left(y_i^\delta\right)-\frac{1}{T_0'\left(y_i^0\right)}w'\left(y_i^0\right)}{\delta}$$
$$=\lim_{\delta\to 0}\frac{\frac{1}{T_\delta'\left(y_i^\delta\right)}w'\left(y_i^\delta\right)-\frac{1}{T_\delta'\left(y_i^\delta\right)}w'\left(y_i^0\right)}{\delta}$$
$$+\lim_{\delta\to 0}\frac{\frac{1}{T_\delta'\left(y_i^\delta\right)}w'\left(y_i^0\right)-\frac{1}{T_0'\left(y_i^0\right)}w'\left(y_i^0\right)}{\delta}. \tag{70}$$

To handle the first summand in (70) we remark that by Taylor expansion with first-order Lagrange remainder we can obtain

$$w'\left(y_i^\delta\right)=\left[w'\left(y_i^0\right)+w''\left(y_i^0\right)\left(y_i^\delta-y_i^0\right)\right]+\left(w''\left(\xi\right)-w''\left(y_i^0\right)\right)\left(y_i^\delta-y_i^0\right).$$

with ξ between y_i^0 and y_i^δ. By the uniform continuity of w'' we get

$$\left(w''\left(\xi\right)-w''\left(y_i^0\right)\right)\to 0$$

as δ goes to 0 and then

$$\left(w''\left(\xi\right)-w''\left(y_i^0\right)\right)\left(y_i^\delta-y_i^0\right)=o\left(\delta\right).$$

uniformly on the circle, then by lemma 19 again

$$w'\left(y_i^\delta\right)=w'\left(y_i^0\right)+w''\left(y_i^0\right)\delta\left[-\frac{\dot{T}\left(y_i^0\right)}{T_0'\left(y_i^0\right)}\right]+h_2\left(x,\delta\right)$$

with

$$||\frac{h_2\left(x,\delta\right)}{\delta}||_{C^0}\to 0.$$

By this we also have that $w(y_i^\delta)\to w(y_i^0)$ in C^1.

Thus we get

$$\lim_{\delta\to 0}\frac{\frac{1}{T_\delta'\left(y_i^\delta\right)}w'\left(y_i^\delta\right)-\frac{1}{T_\delta'\left(y_i^\delta\right)}w'\left(y_i^0\right)}{\delta}=\lim_{\delta\to 0}\frac{1}{T_\delta'\left(y_i^\delta\right)}\frac{w'\left(y_i^\delta\right)-w'\left(y_i^0\right)}{\delta}$$
$$=\lim_{\delta\to 0}\frac{1}{T_\delta'\left(y_i^\delta\right)}\frac{w''\left(y_i^0\right)\delta\left[-\frac{\dot T\left(y_i^0\right)}{T_0'\left(y_i^0\right)}\right]+h_2\left(x,\delta\right)}{\delta}$$
$$=\frac{-w''\left(y_i^0\right)\dot T\left(y_i^0\right)T_0'\left(y_i^0\right)}{T_0'\left(y_i^0\right)^3}\tag{71}$$

in the C^0 topology. The second summand in (70) is estimated as in the proof of lemma (19) (see the calculations from (57) to (58)) obtaining

$$\left[\frac{1}{T_\delta'\left(y_i^\delta\left(x\right)\right)}-\frac{1}{T_0'\left(y_i^0\left(x\right)\right)}\right]=-\frac{y_i^0{}'\dot T'\left(y_i^0\right)T_0'\left(y_i^0\right)-y_i^0{}'T_0''\left(y_i^0\right)\dot T\left(y_i^0\right)}{\left(T_0'\left(y_i^0\right)\right)^2}+O_{C^0}\left(\delta^2\right).$$

Then

$$\lim_{\delta\to 0}\frac{\frac{1}{T_\delta'\left(y_i^\delta\right)}w'\left(y_i^0\right)-\frac{1}{T_0'\left(y_i^0\right)}w'\left(y_i^0\right)}{\delta}=-w'\left(y_i^0\right)\frac{y_i^0{}'\dot T'\left(y_i^0\right)T_0'\left(y_i^0\right)-y_i^0{}'T_0''\left(y_i^0\right)\dot T\left(y_i^0\right)}{\left(T_0'\left(y_i^0\right)\right)^2}.\tag{72}$$

Thus putting together (71) and (72) recalling that $(y_i^0)'=\frac{1}{T_0'(y_i^0)}$, (69) is proved. Thus we have proved the C^1 convergence of the limit (67). This proves that (II) converges in the C^1 topology to the second summand of (46).

Now we estimate the L^1 uniform convergence of term (II) to prove the corresponding part of (48). We now suppose $w\in W^{2,1}$. Using the Taylor formula with integral remainder we have that

$$w\left(y_i^\delta\left(x\right)\right)=w\left(y_i^0\left(x\right)+y_i^\delta\left(x\right)-y_i^0\left(x\right)\right)$$
$$=w\left(y_i^0\left(x\right)\right)+w'\left(y_i^0\left(x\right)\right)\left[y_i^\delta\left(x\right)-y_i^0\left(x\right)\right]+\int_{y_i^0\left(x\right)}^{y_i^\delta\left(x\right)}w''\left(t\right)\left(y_i^\delta\left(x\right)-t\right)\mathrm{d}t.\tag{73}$$

Let us estimate the L^1 norm of the function $x\to\int_{y_i^0(x)}^{y_i^\delta(x)}w''(t)(y_i^\delta(x)-t)\mathrm{d}t$

$$\int_{S^1}\left|\int_{y_i^0\left(x\right)}^{y_i^\delta\left(x\right)}w''\left(t\right)\left(y_i^\delta\left(x\right)-t\right)\mathrm{d}t\right|\mathrm{d}x\leqslant\int_{S^1}\int_{\min\left(y_i^0\left(x\right),y_i^\delta\left(x\right)\right)}^{\max\left(y_i^0\left(x\right),y_i^\delta\left(x\right)\right)}|w''\left(t\right)|\cdot|\left(y_i^\delta\left(x\right)-t\right)|\mathrm{d}t\mathrm{d}x$$

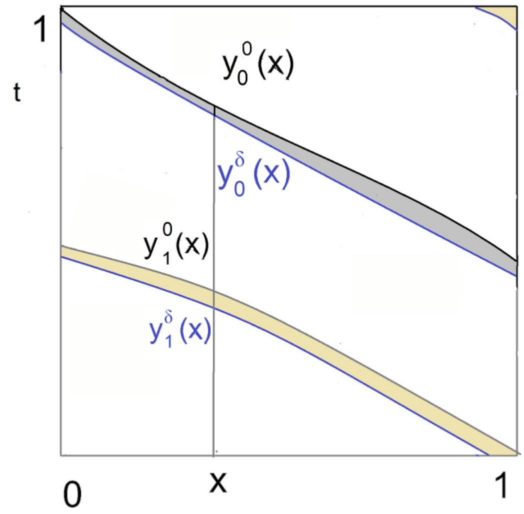

Figure 7. An example of the integration regions considered in (74). The grey region delimited by the inverse branches y_0^0, y_0^δ is the region on which the integral in the first term of (74) is defined, the union of the grey and the yellow region is the region (delimited by all inverse branches of the map before and after perturbation) on which which the integral the last term of (74) is defined.

we change the order of integration and get (see figure 7)

$$\int_{S^1}\int_{\min\left(y_i^0(x),y_i^\delta(x)\right)}^{\max\left(y_i^0(x),y_i^\delta(x)\right)} |w''(t)|\left|\left(y_i^\delta(x)-t\right)\right|\mathrm{d}t\mathrm{d}x \leqslant \int_{S^1}\int_{\min(T_0(t),T_\delta(t))}^{\max(T_0(t),T_\delta(t))} |w''(t)|\left|\left(y_i^\delta(x)-t\right)\right|\mathrm{d}x\mathrm{d}t \tag{74}$$

$$= \int_{S^1}|w''(t)|\int_{\min(T_0(t),T_\delta(t))}^{\max(T_0(t),T_\delta(t))}\left|\left(y_i^\delta(x)-t\right)\right|\mathrm{d}x\mathrm{d}t.$$

Since $\delta \mapsto y_i^\delta(x)$ is uniformly Lipschitz and $|(y_i^\delta(x)-t)| = 0$ at one of the two endpoints of the integration interval $I_\delta(t) := [\min(T_0(t),T_\delta(t)),\max(T_0(t),T_\delta(t))]$ we get that there is K_1 such that $\forall t$, $\max_{x\in I_\delta(t)}|(y_i^\delta(x)-t)| \leqslant K_1\delta$ and that there is K_2 such that $\forall t \int_{\min(T_0(t),T_\delta(t))}^{\max(T_0(t),T_\delta(t))}|(y_i^\delta(x)-t)|\mathrm{d}x \leqslant K_2\delta^2$. Thus

$$\int_{S^1}\left|\int_{y_i^0(x)}^{y_i^\delta(x)} w''(t)\left(y_i^\delta(x)-t\right)\mathrm{d}t\right|\mathrm{d}x \leqslant K_2\delta^2\int_{S^1}|w''(t)|\,\mathrm{d}t \leqslant K_2\delta^2\|w\|_{W^{2,1}}. \tag{75}$$

Hence inserting the estimate from (75) into (73) we obtain

$$w\left(y_i^\delta\right) = w\left(y_i^0\right) - \delta w'\left(y_i^0\right)\frac{\dot{T}\left(y_i^0\right)}{T_0'\left(y_i^0\right)} + O_{L^1}\left(\delta^2\right)$$

where similarly as before we write $g_\delta = O_{L^1}(\delta^2)$ when $\sup_{\delta\in[0,\bar{\delta})} \frac{||g_\delta||_{L^1}}{\delta^2} < +\infty$. We remark that the above $O_{L^1}(\delta^2)$ term depends only on w through its $W^{2,1}$ norm, hence it is uniform as w ranges in the unit ball of such Sobolev space.

Continuing, we write

$$(II) = \frac{1}{\delta}\sum_{i=1}^{d}\frac{w\left(y_i^\delta\right)-w\left(y_i^0\right)}{T_0'\left(y_i^\delta\right)} = -\frac{1}{\delta}\sum_{i=1}^{d}\frac{\delta w'\left(y_i^0\right)\frac{\dot{T}(y_i^0)}{T_0'(y_i^0)}+O_{L^1}\left(\delta^2\right)}{T_0'\left(y_i^\delta\right)}$$
$$= -\sum_{i=1}^{d}\frac{\dot{T}\left(y_i^0\right)w'\left(y_i^0\right)}{T_0'\left(y_i^0\right)T_0'\left(y_i^\delta\right)}+\frac{O_{L^1}\left(\delta^2\right)}{\delta}. \tag{76}$$

Now we recall that by lemma 19, $y_i^\delta = y_i^0 + \delta\left(-\frac{\dot{T}(y_i^0)}{T_0'(y_i^0)}\right) + O_{C^1}(\delta^2)$, furthermore $\frac{\dot{T}(\cdot)}{T_0'(\cdot)T_0'(\cdot)}$ is Lipschitz, thus by (60) we get that for all $1\leqslant i\leqslant d$ as $\delta\to 0$

$$||\frac{\dot{T}\left(y_i^0\right)}{T_0'\left(y_i^0\right)T_0'\left(y_i^\delta\right)}-\frac{\dot{T}\left(y_i^0\right)}{T_0'\left(y_i^0\right)T_0'\left(y_i^0\right)}||_{C^0} = O(\delta).$$

Since w is in $W^{2,1}$, by the Sobolev embedding theorem w' is continuous and there is $C\geqslant 0$ such that $||w'||_{C^0}\leqslant C||w||_{W^{2,1}}$. Thus there is a $C_2\geqslant 0$ such that for sufficiently small δ, for each $1\leqslant i\leqslant d$, one has

$$\sup_{w\in B_{W^{2,1}}(0,1)}||\frac{\dot{T}\left(y_i^0\right)w'\left(y_i^0\right)}{T_0'\left(y_i^0\right)T_0'\left(y_i^\delta\right)}-\frac{\dot{T}\left(y_i^0\right)w'\left(y_i^0\right)}{T_0'\left(y_i^0\right)T_0'\left(y_i^0\right)}||_{C^0}\leqslant C_2\delta,$$

and then

$$\sup_{w\in B_{W^{2,1}}(0,1)}||\sum_{i=1}^{d}\frac{\dot{T}\left(y_i^0\right)w'\left(y_i^0\right)}{T_0'\left(y_i^0\right)T_0'\left(y_i^\delta\right)}-\sum_{i=1}^{d}\frac{\dot{T}\left(y_i^0\right)w'\left(y_i^0\right)}{T_0'\left(y_i^0\right)T_0'\left(y_i^0\right)}||_{C^0}\leqslant dC_2\delta. \tag{77}$$

By (76) and (77) we get

$$\sup_{w\in B_{W^{2,1}}(0,1)}||\frac{1}{\delta}\sum_{i=1}^{d}\frac{w\left(y_i^\delta\right)-w\left(y_i^0\right)}{T_0'\left(y_i^\delta\right)}+L_0\left(\frac{\dot{T}w'}{T_0'}\right)||_{L^1} = O(\delta)$$

giving a uniform estimate as w ranges in the unit ball of $W^{2,1}$.

Term (III): Finally, for the third term (*III*) by lemma 19 we can write

$$T_0'\left(y_i^\delta\right) = T_0'\left(y_i^0\right)+T_0''\left(y_i^0\right)\left(-\frac{\dot{T}\left(y_i^0\right)}{T_0'\left(y_i^0\right)}\right)\delta+O_{C^1}\left(\delta^2\right).$$

Again using lemma 19 we get

$$
\begin{aligned}
(III) &= \frac{1}{\delta}\left(\sum_{i=1}^{d} w\left(y_i^0\right)\left(\frac{1}{T_0'\left(y_i^\delta\right)}-\frac{1}{T_0'\left(y_i^0\right)}\right)\right)\\
&= \frac{1}{\delta}\left(\sum_{i=1}^{d} w\left(y_i^0\right)\left(\frac{T_0'\left(y_i^0\right)-T_0'\left(y_i^\delta\right)}{T_0'\left(y_i^\delta\right)T_0'\left(y_i^0\right)}\right)\right)\\
&= \frac{1}{\delta}\left(\sum_{i=1}^{d} w\left(y_i^0\right)\left(\frac{T_0'\left(y_i^0\right)-\left(T_0'\left(y_i^0\right)+T_0''\left(y_i^0\right)\left(-\frac{\dot{T}(y_i^0)}{T_0'(y_i^0)}\right)\delta+O_{C^1}\left(\delta^2\right)\right)}{T_0'\left(y_i^\delta\right)T_0'\left(y_i^0\right)}\right)\right)\\
&= \left(\sum_{i=1}^{d} w(y_i^0)\left(\frac{\dot{T}(y_i^0)T_0''(y_i^0)}{T_0'^2(y_i^0)T_0'(y_i^\delta)}\right)\right)+\frac{O_{C^1}(\delta^2)}{\delta}.
\end{aligned}
$$

Finally, by lemma 19 and (59), similarly to what was done for (64), we get

$$
\lim_{\delta\to 0}\frac{1}{\delta}\left(\sum_{i=1}^{d} w\left(y_i^0\right)\left(\frac{1}{T_0'\left(y_i^\delta\right)}-\frac{1}{T_0'\left(y_i^0\right)}\right)\right)=L_0\left(\frac{\dot{T}T_0''}{T_0'^2}w\right)
$$

in C^1. This proves that (*III*) converges in the C^1 topology to the third summand of (46).

Furthermore, again with a reasoning similar to what done for (76) (see the lines from (76) to (77)) by lemma 19 and (60), and using arguments similar to those regarding uniformity of constants for Term (I) after equation (66), we obtain

$$
\left\|\frac{1}{\delta}\left(\sum_{i=1}^{d} w\left(y_i^0\right)\left(\frac{1}{T_0'\left(y_i^\delta\right)}-\frac{1}{T_0'\left(y_i^0\right)}\right)\right)-L_0\left(\frac{\dot{T}T_0''}{T_0'^2}w\right)\right\|_{C^0}=O(\delta),
$$

uniformly for $w\in B_{W^{2,1}}(0,1)$.

□

A.5. Proof of proposition 3

In the following Lemma, we establish more precise regularity for the eigenvectors of the simple eigenvalues of the transfer operator.

Lemma 25. *Let $L_0 : W^{1,1}\to W^{1,1}$ be the transfer operator associated to an expanding map $T_0\in C^4(S^1)$. Let λ_0 be a simple eigenvalue of L_0 such that $|\lambda_0|>\frac{1}{\inf(|T_0'|)}$, let $v_0\in W^{1,1}(S^1)$ be the normalised eigenvector associated to λ_0, as in (III) of proposition 3, then $v_0\in W^{3,1}(S^1)$.*

Proof. Lemma A.3 of [4] (whose statement is recalled in the proof of Proposition 15) can be applied to the transfer operator $L_0 : W^{1,1}\to W^{1,1}$, obtaining that $v_0\in W^{2,1}$ as follows.

By lemma 18 we get that L_0 satisfies Lasota–Yorke inequalities both with $W^{1,1}$ and L^1 and with $W^{2,1}$ and $W^{1,1}$ as a weak and strong spaces. This implies that $L_0 : W^{1,1}\to W^{1,1}$ is continuous and preserves $W^{2,1}$. Furthermore the restriction of L_0 to $W^{2,1}$ is also continuous as an operator on $W^{2,1}$. It is also well known that by the Lasota–Yorke inequalities (lemma 18) and the compact immersion between the strong and the weak spaces provided by the Rellich–Kondracov theorem, the essential spectral radii of $L_0 : W^{1,1} \to W^{1,1}$ and $L_0 : W^{2,1} \to W^{2,1}$ are smaller than $\frac{1}{\inf(|T_0'|)}$ (see e.g. lemma 2.2 of [6]). Thus lemma A.3 of [4] can be applied, giving that the generator v_0 of the one dimensional eigenspace associated to λ_0 is contained in $W^{2,1}$. Now, since for a C^4 map, lemma 18 also gives a $W^{3,1}$, $W^{2,1}$ Lasota Yorke inequality, we can repeat the same reasoning using $W^{3,1}$, $W^{2,1}$ as a strong and weak space, obtaining that $v_0 \in W^{3,1}$. □

Proof of proposition 3. Below we set B_{ss}, B_s, B_w to be $W^{2,1}$, $W^{1,1}$, L^1, respectively.

Proof of part I: the linear response formula for the invariant density (4) will follow as a direct consequence of theorem 12 and Proposition 22.

To show that theorem 12 can be applied we show that its assumptions (25)–(27) are verified for our maps and perturbations. Equation (25) will follow from proposition 22. Indeed, by (A0), $T_0 \in C^4$, hence by lemma 25 the normalised eigenvector v_0 of λ_0 is contained in $W^{3,1}$ and hence is contained in C^2.

Furthermore, propositions 19 and 21 provide the existence and stability of the resolvent as required by the assumptions (26) and (27), respectively, of theorem 12.

Proof of part II: the linear response formula (5) for isolated eigenvalues will be established by applying proposition 16. Again, the necessary properties of the operator in this case are established in proposition 22. To apply proposition 16 we verify that the assumptions (25) and (GL1)–(GL7) hold for our maps and perturbations. Assumption (25) has been established above in the proof of Part I. The assumptions (GL1), (GL2) and (GL3) are verifed by the Lasota–Yorke inequalities established in lemma 18. The small perturbation assumptions needed at (GL4) are established in proposition 20. Next we verify that (GL5)–(GL7) are satisfied for perturbations of expanding maps satisfying (A0), (A1) and (A2).

(GL5): we consider the derivative operator $\dot{L}$ as defined in (3). Since we consider the stronger and weaker spaces B_{ss}, B_s, B_w to be $W^{2,1}$, $W^{1,1}$, L^1, from (3) we directly verify that $\dot{L} : W^{1,1} \to L^1$ and $\dot{L} : W^{3,1} \to W^{1,1}$ are bounded operators.

(GL6): We must show that $||L_\delta - L_0 - \delta\dot{L}||_{W^{2,1}\to L^1} \leqslant C\delta^2$. This follows by multiplying (48) by δ.

(GL7): by the Rellich–Kondracov theorems there are compact immersions between the spaces $W^{2,1}, W^{1,1}$, and L^1 and hence the assumption (GL7) is satisfied. □

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